ABRASIVES (MANUFACTURED)

(Fused aluminum oxide, silicon carbide, and metallic abrasives)
(Data in metric tons unless otherwise specified)

Domestic Production and Use: In 2024, fused aluminum oxide was produced by two companies at three plants in the United States and Canada. Production of crude fused aluminum oxide had an estimated value of $3.9 million. Silicon carbide was produced by two companies at two plants in the United States. Production of crude silicon carbide had an estimated value of about $25 million. Metallic abrasives were produced by 11 companies in eight States. Production of metallic abrasives had an estimated value of about $160 million, and metallic abrasive shipments were valued at $310 million. Bonded and coated abrasive products accounted for most abrasive uses of fused aluminum oxide and silicon carbide. Metallic abrasives are used primarily for steel shot and grit and cut wire shot, which are used for sandblasting, peening, and stonecutting applications.

Salient Statistics—United States:	**2020**	**2021**	**2022**	**2023**	**2024[e]**
Production:					
Fused aluminum oxide, crude[1, 2]	10,000	10,000	20,000	25,000	25,000
Silicon carbide[2]	35,000	35,000	40,000	45,000	45,000
Metallic abrasives	176,000	176,000	180,000	198,000	200,000
Shipments, metallic abrasives	194,000	193,000	199,000	227,000	230,000
Imports for consumption:					
Fused aluminum oxide	121,000	159,000	225,000	120,000	120,000
Silicon carbide	88,400	125,000	165,000	114,000	110,000
Metallic abrasives	25,800	26,400	20,100	17,800	17,000
Exports:					
Fused aluminum oxide	11,400	13,500	14,400	9,540	9,400
Silicon carbide	8,310	12,000	12,000	10,100	11,000
Metallic abrasives	18,000	20,100	23,900	24,100	20,000
Consumption, apparent:					
Fused aluminum oxide[3]	109,000	146,000	210,000	110,000	110,000
Silicon carbide[4]	115,000	148,000	193,000	149,000	140,000
Metallic abrasives[5]	202,000	199,000	195,000	220,000	230,000
Price, average unit value of imports, dollars per metric ton:					
Fused aluminum oxide, crude	666	674	797	655	620
Fused aluminum oxide, ground and refined	1,180	1,290	1,560	1,380	1,500
Silicon carbide, crude	628	587	1,080	905	770
Metallic abrasives	1,130	1,510	2,130	1,850	2,000
Net import reliance[6] as a percentage of apparent consumption:					
Fused aluminum oxide	>95	>95	>95	>95	>95
Silicon carbide	70	76	79	70	69
Metallic abrasives	4	3	E	E	E

Recycling: Up to 30% of fused aluminum oxide may be recycled, and about 5% of silicon carbide is recycled.

Import Sources (2020–23): Fused aluminum oxide, crude: China, 91%; and other, 9%. Fused aluminum oxide, ground and refined: Canada, 28%; Brazil, 19%; China, 15%; Austria, 14%; and other, 24%. Total fused aluminum oxide: China, 64%; Canada, 11%; Brazil, 7%; Austria, 5%; and other, 13%. Silicon carbide, crude: China, 94%; and other, 6%. Silicon carbide, ground and refined: China, 58%; Brazil, 16%; Canada, 11%; Norway, 8%; and other, 7%. Total silicon carbide: China, 85%; Brazil, 4%; Canada, 3%; and other, 8%. Metallic abrasives: Canada, 47%; Turkey, 12%; Thailand, 9%; Japan, 7%; and other, 25%.

Tariff: Item	**Number**	**Normal Trade Relations 12–31–24**
Artificial corundum, crude	2818.10.1000	Free.
White, pink, ruby artificial corundum, greater than 97.5% aluminum oxide, grain	2818.10.2010	1.3% ad valorem.
Artificial corundum, not elsewhere specified or included, fused aluminum oxide, grain	2818.10.2090	1.3% ad valorem.
Silicon carbide, crude	2849.20.1000	Free.
Silicon carbide, grain	2849.20.2000	0.5% ad valorem.
Iron, pig iron, or steel granules	7205.10.0000	Free.

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ABRASIVES (MANUFACTURED)

Depletion Allowance: None.

Government Stockpile: None.

Events, Trends, and Issues: In 2024, China was the world's leading manufacturer of abrasive fused aluminum oxide and abrasive silicon carbide. Imports from China, where production costs were lower, continued to challenge abrasives manufacturers in the United States and Canada. China accounted for 94% of United States imports of crude fused aluminum oxide, 14% of ground and refined fused aluminum oxide imports, 98% of crude silicon carbide imports, and 67% of ground and refined silicon carbide imports. Foreign competition was expected to persist and continue to limit production in North America. The import quantities of abrasive silicon carbide (crude and ground and refined) in 2024 were 5% less and 13% more, respectively, than those in 2023.

The United States was a net exporter of metallic abrasives in 2024, 2023, and 2022 compared with being a net importer in 2021 and 2020. The import quantity of metallic abrasives in 2024 was 6% less than that in 2023. Canada was the leading supplier of metallic abrasive imports.

The consumption of abrasives in the United States is influenced by activity in the manufacturing sectors that use them, particularly the aerospace, automotive, furniture, housing, and steel industries. The U.S. abrasive markets also are influenced by technological trends.

World Production Capacity:

	Fused aluminum oxide[e]		Silicon carbide[e]	
	2023	2024	2023	2024
United States	—	—	40,000	40,000
United States and Canada	60,000	60,000	—	—
Australia	50,000	50,000	—	—
Austria	90,000	90,000	—	—
Brazil	50,000	50,000	40,000	40,000
China	800,000	800,000	450,000	450,000
France	40,000	40,000	20,000	20,000
Germany	80,000	80,000	35,000	35,000
India	40,000	40,000	5,000	5,000
Japan	15,000	15,000	60,000	60,000
Mexico	—	—	45,000	45,000
Norway	—	—	80,000	80,000
Venezuela	—	—	30,000	30,000
Other countries	80,000	80,000	200,000	200,000
World total (rounded)	1,310,000	1,310,000	1,010,000	1,010,000

World Resources:[7] Although domestic resources of raw materials for fused aluminum oxide production are limited, adequate resources are available in the Western Hemisphere. Domestic resources are more than adequate for silicon carbide production.

Substitutes: Natural and manufactured abrasives, such as garnet, emery, metallic abrasives, or staurolite, can be substituted for fused aluminum oxide and silicon carbide in various applications.

[e]Estimated. E Net exporter. — Zero.

[1]Production data for fused aluminum oxide are combined data from the United States and Canada to avoid disclosing company proprietary data.

[2]Rounded to the nearest 5,000 tons to avoid disclosing company proprietary data.

[3]Defined as imports – exports because production includes data from Canada; actual consumption is higher than that shown.

[4]Defined as production + imports – exports.

[5]Defined as shipments + imports – exports.

[6]Defined as imports – exports.

[7]See Appendix C for resource and reserve definitions and information concerning data sources.

ALUMINUM[1]

(Data in thousand metric tons unless otherwise specified)

Domestic Production and Use: In 2024, two companies operated four primary aluminum smelters in four States. Two of these smelters operated at full capacity throughout the year, whereas two smelters operated at reduced capacity. One smelter located in Hawesville, KY, has been temporarily shut down since 2022, and another smelter in New Madrid, MO, was temporarily shut down in January. Domestic smelter capacity was 1.36 million tons per year in 2024, unchanged from that in 2023. Estimated primary production decreased by 11% from that in 2023, whereas estimated secondary production from new and old scrap was 5% more than that in 2023. Based on published prices, the value of primary aluminum production was about $1.9 billion, 9% less than that in 2023. The estimated average annual U.S. market price increased by 3% from that in 2023. Transportation applications accounted for 36% of domestic consumption; the remainder was used in packaging, 23%; building, 14%; electrical, 9%; consumer durables and machinery, 8% each; and other, 2%.

Salient Statistics—United States:	2020	2021	2022	2023	2024[e]
Production:					
Primary	1,010	889	861	750	670
Secondary (from old scrap)	1,420	1,520	1,480	1,560	1,600
Secondary (from new scrap)	1,630	1,780	1,920	1,870	2,000
Imports for consumption:					
Crude and semimanufactures	4,260	4,820	5,710	4,890	4,800
Scrap	542	679	685	677	660
Exports:					
Crude and semimanufactures	906	900	1,040	1,230	1,400
Scrap	1,840	2,100	2,000	2,060	1,600
Consumption, apparent[2]	3,930	4,020	4,890	4,150	4,300
Supply, apparent[3]	5,560	5,800	6,810	6,010	6,300
Price, ingot, average U.S. market (spot), cents per pound[4]	89.7	138.5	152.6	125.9	130
Stocks, yearend:					
Aluminum industry	1,490	1,870	2,050	1,820	1,600
London Metal Exchange (LME), U.S. warehouses[5]	235	69	9	5	10
Employment, number[6]	30,100	28,900	30,200	30,500	30,000
Net import reliance[7] as a percentage of apparent consumption	38	40	52	44	47

Recycling: In 2024, aluminum recovered from purchased scrap in the United States was about 3.6 million tons, of which about 56% came from new scrap (manufacturing) and 44% from old scrap (discarded aluminum products). Aluminum recovered from old scrap was equivalent to about 37% of apparent consumption.

Import Sources (2020–23): Canada, 56%; United Arab Emirates, 8%; Bahrain, 4%; China,[8] 3%; and other, 29%.

Tariff: Item	Number	Normal Trade Relations 12–31–24
Aluminum, not alloyed:		
Unwrought (in coils)	7601.10.3000	2.6% ad valorem.
Unwrought (greater than 99.8% aluminum)	7601.10.6030	Free.
Aluminum alloys, unwrought (billet)	7601.20.9045	Free.
Aluminum scrap:		
Used beverage container scrap	7602.00.0035	Free.
Industrial process scrap	7602.00.0095	Free.
Other	7602.00.0097	Free.

Depletion Allowance: Not applicable.[1]

Government Stockpile:[9]

	FY 2024		FY 2025	
Material	Potential acquisitions	Potential disposals	Potential acquisitions	Potential disposals
Aluminum, high-purity and alloys	18.5	—	3.2	—

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ALUMINUM

Events, Trends, and Issues: In January 2024, a 263,000-ton-per-year primary aluminum smelter in New Madrid, MO, shut down its full production with no scheduled restart. In June, a Republic of Korea-based auto parts manufacturer opened a $128 million production and casting facility in Tuskegee, AL. The facility produced aluminum components for an automobile manufacturing plant in Montgomery, AL. In September, construction began on a $40 million aluminum recycling facility in Russellville, KY, which will process dross and scrap from a nearby aluminum casting plant. Also in September, a United Arab Emirates-based aluminum producer announced the acquisition of a 110,000-ton-per-year United States-based secondary aluminum producer located in Rosemont, MN.

In March, the U.S. Department of Energy announced grants to four aluminum operations, including $500 million to build a new aluminum smelter within the Ohio River and Mississippi River Basins; $75 million to build a low-carbon aluminum casting plant in Ravenswood, WV; $67.3 million to build a low-waste recycling facility in Wabash, IN; and $22.3 million to upgrade casting and rolling equipment at a casting and rolling mill in Fort Lupton, CO. In July, the U.S. Department of Defense awarded $23 million to increase aluminum casting capacity by as much as 136,000 tons per year at an aluminum rolling facility in Muscle Shoals, AL.

In April, the United States coordinated with the United Kingdom to ban imports of aluminum from Russia into both countries and to restrict the sale of these metals on global metal exchanges and in over-the-counter derivative trading. In July, the United States imposed 10% duties on imports of aluminum products and derivative aluminum products from Mexico that contain primary aluminum for which the primary or secondary country of smelt or the most recent country of cast was Belarus, China, Iran, or Russia. In September, the United States increased tariffs on aluminum products imported from China, from 0%–7.5% to 25%.

World Smelter Production and Capacity: Capacity data for China, India, and other countries were revised based on company and Government reports.

	Smelter production		Yearend capacity	
	2023	2024[e]	2023	2024[e]
United States	750	670	1,360	1,360
Australia	1,560	1,500	1,730	1,730
Bahrain	1,620	1,600	1,600	1,600
Brazil	1,020	1,100	1,280	1,280
Canada	[e]3,200	3,300	3,270	3,270
China	41,600	43,000	44,400	44,700
Iceland	[e]770	780	880	880
India	[e]4,100	4,200	4,100	4,200
Malaysia	[e]940	870	1,080	1,080
Norway	[e]1,300	1,300	1,460	1,460
Russia	[e]3,700	3,800	4,080	4,080
United Arab Emirates	2,660	2,700	2,790	2,790
Other countries	6,780	6,800	10,300	10,000
World total (rounded)	70,000	72,000	78,300	78,400

World Resources:[10] Global resources of bauxite are estimated to be between 55 billion and 75 billion tons and are sufficient to meet world demand for aluminum metal well into the future.

Substitutes: Composites can substitute for aluminum in aircraft fuselages and wings. Glass, paper, plastics, and steel can substitute for aluminum in packaging. Composites, magnesium, steel, and titanium can substitute for aluminum in ground transportation uses. Composites, steel, vinyl, and wood can substitute for aluminum in construction. Copper can replace aluminum in electrical and heat-exchange applications.

[e]Estimated. — Zero.

[1]See also the Bauxite and Alumina chapter.

[2]Defined as primary production + secondary production from old scrap + imports – exports ± adjustments for stock changes; excludes imported scrap.

[3]Defined as primary production + secondary production + imports – exports ± adjustments for stock changes; excludes imported scrap.

[4]Source: S&P Global Platts Metals Week.

[5]Includes off-warrant stocks of primary and alloyed aluminum.

[6]Alumina and aluminum production workers (North American Industry Classification System—3313). Source: U.S. Department of Labor, Bureau of Labor Statistics.

[7]Defined as imports – exports ± adjustments for industry stock changes; excludes imported scrap.

[8]Includes Hong Kong.

[9]See Appendix B for definitions.

[10]See Appendix C for resource and reserve definitions and information concerning data sources.

ANTIMONY

(Data in metric tons, antimony content, unless otherwise specified)

Domestic Production and Use: In 2024, no marketable antimony was mined in the United States. Primary antimony metal and oxide were produced by one company in Montana using imported feedstock; data were not available. Secondary antimony production came from antimonial lead recovered from spent lead-acid batteries and was intended for the lead-acid battery industry. The estimated value of secondary antimony produced in 2024 was about $73 million. Recycling supplied about 15% of estimated domestic apparent consumption, and the remainder came from imports. In the United States, the leading uses of antimony were metal products, including antimonial lead and ammunition, 40%; flame retardants, 39%; and nonmetal products, including ceramics and glass and rubber products, 21%.

Salient Statistics—United States:	2020	2021	2022	2023	2024[e]
Production:					
Mine (recoverable antimony)	—	—	—	—	—
Smelter:					
Primary	254	19	W	W	W
Secondary	3,520	4,050	4,100	3,490	3,500
Imports for consumption:					
Ore and concentrates	105	31	29	6	310
Oxide	15,000	19,100	16,900	14,000	20,000
Unwrought, powder	5,200	6,970	6,510	6,060	4,100
Antimony articles[1]	318	514	1,790	1,620	330
Waste and scrap[1]	6	13	71	3	17
Exports:					
Ore and concentrates[1]	10	9	53	24	—
Oxide	1,230	1,530	2,420	1,740	2,200
Unwrought, powder	269	824	1,230	1,510	1,500
Antimony articles[1]	97	97	585	433	79
Waste and scrap[1]	11	136	26	2	53
Consumption, apparent[2]	22,400	27,800	23,900	20,300	24,000
Price, metal, average, dollars per pound[3]	2.67	5.31	6.18	5.49	9.50
Net import reliance[4] as a percentage of apparent consumption	83	85	83	83	85

Recycling: The bulk of secondary antimony is recovered at secondary lead smelters as antimonial lead, most of which was generated by, and then consumed by, the lead-acid battery industry.

Import Sources (2020–2023): Ore and concentrates: Italy, 44%; China, 23%; Belgium, 16%; India, 10%; and other, 7%. Oxide: China,[5] 76%; Belgium, 11%; Bolivia, 6%; and other, 7%. Unwrought metal and powder: India, 25%; China,[5] 24%; Thailand, 13%; Vietnam, 12%; and other, 26%. Total metal and oxide: China,[5] 63%; Belgium, 8%, India, 6%; Bolivia, 5%; and other, 18%.

Tariff: Item	Number	Normal Trade Relations 12–31–24
Ore and concentrates	2617.10.0000	Free.
Antimony oxide	2825.80.0000	Free.
Unwrought antimony; powders	8110.10.0000	Free.
Waste and scrap	8110.20.0000	Free.
Antimony articles	8110.90.0000	Free.

Depletion Allowance: 22% (domestic), 14% (foreign).

Government Stockpile:[6]

	FY 2024		FY 2025	
Material	Potential acquisitions	Potential disposals	Potential acquisitions	Potential disposals
Antimony	1,100	—	700	—

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ANTIMONY

Events, Trends, and Issues: In August 2024, China's Government announced that companies would need to apply for export licenses to export antimony ore, metals, oxides, hydrides, indium antimonides, organo-antimony compounds, and gold-antimony separation technology. After that announcement, the antimony metal price nearly doubled, from $8.91 per pound in July to $17.50 per pound in November, according to Argus Media Group. In December 2024, China banned all exports of antimony to the United States.

In February, a mining company in Idaho was conditionally awarded additional funding from the U.S. Department of Defense to reestablish a domestic source of antimony, bringing its total Department of Defense funding to $59.4 million. According to the company, the project has total proven and probable mineral reserves of 14.2 million tons of 0.42% antimony ore.

World Mine Production and Reserves: Reserves for China and Vietnam were revised based on Government reports.

	Mine production		Reserves[7]
	2023	2024[e]	
United States	—	—	[8]60,000
Australia	1,860	2,000	[9]140,000
Bolivia	3,700	3,700	310,000
Burma	[e]4,500	4,500	140,000
Canada	—	—	78,000
China	[e]62,300	60,000	670,000
Guatemala	49	50	NA
Iran	[e]500	500	NA
Kazakhstan	[e]40	40	NA
Kyrgyzstan	20	20	260,000
Laos	[e]200	200	NA
Mexico	800	800	18,000
Pakistan	250	250	26,000
Russia	13,000	13,000	350,000
Tajikistan	17,000	17,000	50,000
Turkey	[e]1,600	1,600	99,000
Vietnam	300	300	54,000
World total (rounded)[10]	106,000	100,000	>2,000,000

World Resources:[7] U.S. resources of antimony are mainly in Alaska, Idaho, Montana, and Nevada. Principal identified world resources are in Australia, Bolivia, Burma, China, Mexico, Russia, South Africa, and Tajikistan. Additional antimony resources may occur in Mississippi Valley-type lead deposits in the Eastern United States.

Substitutes: Selected organic compounds and hydrated aluminum oxide are substitutes as flame retardants. Chromium, tin, titanium, zinc, and zirconium compounds substitute for antimony chemicals in enamels, paint, and pigments. Combinations of calcium, copper, selenium, sulfur, and tin are substitutes for alloys in lead-acid batteries.

[e]Estimated. NA Not available. W Withheld to avoid disclosing company proprietary data. — Zero.

[1]Gross weight.

[2]Defined as secondary production from old scrap + imports of antimony in oxide and unwrought metal – exports of antimony in oxide and unwrought metal.

[3]Antimony minimum 99.65%, cost, insurance, and freight. Source: Argus Media Group, Argus Non-Ferrous Markets.

[4]Defined as imports of antimony in oxide and unwrought metal, powder – exports of antimony in oxide and unwrought metal, powder.

[5]Includes Hong Kong.

[6]See Appendix B for definitions.

[7]See Appendix C for resource and reserve definitions and information concerning data sources.

[8]Company-reported probable reserves for the Stibnite Gold Project in Idaho.

[9]For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 20,000 tons.

[10]In addition to the countries listed, antimony may have been produced in other countries, but available information was inadequate to make reliable estimates of output.

ARSENIC

(Data in metric tons, arsenic content,[1] unless otherwise specified)

Domestic Production and Use: Arsenic trioxide and primary arsenic metal have not been produced in the United States since 1985. The principal use for arsenic compounds was in herbicides and insecticides. Arsenic trioxide was predominantly used for the production of arsenic acid, which is a key ingredient in the production of chromated copper arsenate (CCA) preservatives. CCA preservatives are used for the pressure treating of lumber for primarily nonresidential applications such as light poles, marine applications, and retaining walls. Seven companies produced CCA-treated wood in the United States in 2024. High-purity (99.9999%) arsenic metal was used to produce gallium-arsenide (GaAs) semiconductors for solar cells, space research, and telecommunications; germanium-arsenide-selenide specialty optical materials; and indium-gallium-arsenide (InGaAs) for use in shortwave infrared technology. Arsenic metal was used as an antifriction additive for bearings, to harden lead shot and clip-on wheel weights, and to strengthen the grids in lead-acid storage batteries. The estimated value of arsenic compounds and metal imported domestically in 2024 was $11 million. Given that arsenic metal has not been produced domestically since 1985, it is likely that only a small portion of the material reported by the U.S. Census Bureau as arsenic exports was pure arsenic metal, and most of the material that was reported under this category reflects the gross weight of alloys, compounds, residues, scrap, and waste products containing arsenic. Therefore, the estimated consumption reported under U.S. salient statistics reflects only imports of arsenic products. Domestically, the leading uses of arsenic were as follows: herbicides and insecticides and wood preservatives, 84%; semiconductor, 5%; metallurgical, 3%; and other, 8%.

Salient Statistics—United States:	**2020**	**2021**	**2022**	**2023**	**2024[e]**
Imports for consumption:[2]					
Arsenic metal	522	835	934	612	530
Compounds	7,690	4,730	9,190	5,810	[3]8,600
Total	8,220	5,560	10,100	6,430	9,100
Exports, all forms of arsenic (gross weight)	29	31	82	40	[3]570
Consumption, estimated, all forms of arsenic[4]	8,220	5,560	10,100	6,430	9,100
Price, metal, annual average, U.S. warehouse,[5] dollars per pound	1.08	1.11	1.67	2.05	2
Net import reliance[6] as a percentage of estimated consumption, all forms of arsenic	100	100	100	100	100

Recycling: Arsenic metal was contained in new scrap recycled during GaAs semiconductor manufacturing. Arsenic-containing process water was internally recycled at wood treatment plants where CCA was used. Although scrap electronic circuit boards, relays, and switches may contain arsenic, no arsenic was known to have been recovered during the recycling process to recover other contained metals. No arsenic was recovered domestically from arsenic-containing residues and dusts generated at nonferrous smelters in the United States.

Import Sources (2020–23):[2] Arsenic acid: Malaysia, 99%; and other, 1%. Arsenic metal: China,[7] 96%; Japan, 3%; and other, 1%. Arsenic trioxide: China, 58%; Morocco, 34%; Belgium, 5%; and other, 3%. All forms of arsenic: China,[7] 52%; Morocco, 26%; Malaysia, 16%; Belgium, 4%; and other, 2%.

Tariff: Item	**Number**	**Normal Trade Relations 12–31–24**
Arsenic metal	2804.80.0000	Free.
Arsenic acid	2811.19.1000	2.3% ad valorem.
Arsenic trioxide	2811.29.1000	Free.
Arsenic trichloride	2812.19.0010	3.7% ad valorem.
Arsenic sulfide	2813.90.1000	Free.

Depletion Allowance: 14% (domestic and foreign).

Government Stockpile: None.

Events, Trends, and Issues: Peru, China, and Morocco, in descending order of production, continued to be the leading global producers of arsenic trioxide, accounting for about 95% of estimated world production in 2024. China and the Republic of Korea accounted for 92% of United States imports of arsenic trioxide in 2024. China supplied 96% of United States arsenic metal imports through July 2024. Malaysia supplied all the arsenic acid that was imported through July 2024.

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ARSENIC

High-purity arsenic metal was used to produce GaAs, indium-arsenide, and InGaAs semiconductors that were used in aerospace devices, biomedical devices, military applications, mobile devices, optoelectronic devices, photovoltaic applications, satellites, and wireless communications. Total revenues from GaAs devices increased in 2024 because of an increase in the deployment of fifth-generation networks and consumer devices. A variety of GaAs wafer manufacturers ranging from large, multinational corporations to small, privately owned companies competed in this industry, but the top six producers accounted for more than 75% of the global market. See also the Gallium chapter.

World Production and Capacity:

	Production[e, 8] (arsenic trioxide, gross weight)		Refinery capacity (arsenic trioxide, gross weight)[9]
	2023	2024[e]	2024[e]
United States	—	—	—
Belgium	1,000	1,000	1,500
China	24,000	24,000	30,000
Japan[e]	40	40	60
Morocco	6,000	6,000	8,000
Peru	30,000	27,000	37,000
Russia[e]	500	200	4,000
World total (rounded)	61,500	58,000	81,000

World Resources:[10] Arsenic may be obtained from copper, gold, and lead smelter flue dust, as well as from roasting arsenopyrite, the most abundant ore mineral of arsenic. Arsenic has been recovered from orpiment and realgar in China, Peru, and the Philippines and from copper-gold ores in Chile, and arsenic is associated with gold occurrences in Canada. Orpiment and realgar from gold mines in Sichuan Province, China, were stockpiled for later recovery of arsenic. Arsenic also may be recovered from enargite, a copper mineral. Arsenic trioxide was produced at the hydrometallurgical complex of Guemassa, near Marrakech, Morocco, from cobalt-arsenide ore from the Bou Azzer Mine. World reserve data were unavailable but were estimated to be more than 20 times world production.

Substitutes: Substitutes for CCA in wood treatment include alkaline copper quaternary, ammoniacal copper quaternary, ammoniacal copper zinc arsenate, alkaline copper quaternary boron-based preservatives, copper azole, copper citrate, and copper naphthenate. Treated wood substitutes include concrete, plastic composite material, plasticized wood scrap, or steel. Silicon-based complementary metal-oxide semiconductor power amplifiers compete with GaAs power amplifiers in midtier third-generation cellular handsets. Many semiconductor manufacturers were moving away from GaAs- and silicon-based lateral diffused metal-oxide-semiconductor field-effect transistors to those using gallium nitride. Indium phosphide components can be substituted for GaAs-based infrared laser diodes in some specific-wavelength applications, and helium-neon lasers compete with GaAs in visible laser diode applications. Silicon is the principal competitor with GaAs in solar-cell applications. In many defense-related applications, GaAs-based integrated circuits are used because of their unique properties, and no effective substitutes exist for GaAs in these applications. In heterojunction bipolar transistors, GaAs is being replaced in some applications by silicon-germanium.

[e]Estimated. — Zero.

[1]Arsenic content of arsenic metal is 100%; arsenic content of arsenic compounds is 58.2% for arsenic acids, 60.7% for arsenic sulfides, 41.33% for arsenic trichloride, and 75.71% for arsenic trioxide.

[2]Arsenic content calculated from the reported gross weight of imports. See footnote 1 for content percentages of arsenic metal and compounds.

[3]In 2024, includes arsenic trichloride; imports were 450 tons, arsenic content, and exports were 380 tons, gross weight. There were no trade data for arsenic trichloride in previous years.

[4]Estimated to be the same as total imports.

[5]Minimum 99% arsenic. Source: Argus Media Group, Argus Non-Ferrous Markets.

[6]Defined as imports.

[7]Includes Hong Kong.

[8]Includes calculated arsenic trioxide equivalent of output of elemental arsenic compounds other than arsenic trioxide; inclusion of such materials would not duplicate reported arsenic trioxide production. Chile and Mexico were thought to be significant producers of commercial-grade arsenic trioxide but have reported no production in recent years.

[9]Yearend operation capacity.

[10]See Appendix C for resource and reserve definitions and information concerning data sources.

ASBESTOS

(Data in metric tons unless otherwise specified)

Domestic Production and Use: The last U.S. producer of asbestos ceased operations in 2002 as a result of the decline in domestic and international asbestos use associated with health and liability issues. Since then, the United States has been wholly dependent on imports to meet manufacturing needs. All of the unmanufactured asbestos fiber imported into and used within the United States has consisted of chrysotile since no later than 1999. In 2024, domestic consumption of chrysotile was estimated to be 110 tons; all consumption was from stockpiles, as no chrysotile was imported. The chloralkali industry, which uses chrysotile in nonreactive semipermeable diaphragms that prevent chlorine generated at the anode of an electrolytic cell from reacting with sodium hydroxide generated at the cathode, has accounted for 100% of U.S. asbestos fiber consumption since no later than 2015. In addition to unmanufactured asbestos fiber, an unknown quantity of asbestos is imported annually within manufactured products. According to the U.S. Environmental Protection Agency (EPA), the only imported items known to contain asbestos are aftermarket automotive brakes and linings and other vehicle friction products, brake blocks used in the oil industry, and sheet and other gaskets.[1]

Salient Statistics—United States:[2]	2020	2021	2022	2023	2024[e]
Imports for consumption[3]	305	41	224	—	—
Exports[4]	—	—	—	—	—
Consumption, estimated[5]	450	310	290	150	110
Price, average U.S. customs unit value of imports, dollars per ton	2,110	1,880	2,630	NA	NA
Net import reliance[6] as a percentage of estimated consumption	100	100	100	100	100

Recycling: None.

Import Sources (2020–23): Brazil, 91%; and Russia, 9%. The U.S. Census Bureau reported imports from China and Poland during this time period, but bill of lading information, data reported by the Government of China, and an asbestos ban in Poland suggest that these shipments were misclassified.

Tariff: Item	Number	Normal Trade Relations 12–31–24
Crocidolite	2524.10.0000	Free.
Amosite	2524.90.0010	Free.
Chrysotile:		
Crudes	2524.90.0030	Free.
Milled fibers, group 3 grades	2524.90.0040	Free.
Milled fibers, group 4 and 5 grades	2524.90.0045	Free.
Other	2524.90.0055	Free.
Other, asbestos	2524.90.0060	Free.

Depletion Allowance: 22% (domestic), 10% (foreign).

Government Stockpile: None.

Events, Trends, and Issues: Consumption of unmanufactured asbestos fiber in the United States has decreased significantly during the past several decades, from a record high of 803,000 tons in 1973 to 500 tons or less in each year since 2018. This decline has taken place as a result of health and liability issues associated with asbestos use, leading to the displacement of asbestos from traditional domestic markets by substitutes, alternative materials, and new technology. The chloralkali industry is the only remaining U.S. consumer of asbestos in mineral form.

In March 2024, the EPA issued a final rule[1] that will prohibit the commercial use, distribution in commerce, import, manufacturing, and processing of chrysotile for all chrysotile-containing products that are still used in the United States: aftermarket automotive brakes and linings and other vehicle friction products, diaphragms used in the chloralkali industry, oilfield brake blocks, and sheet and other gaskets. Imports of chrysotile for use in the chloralkali industry were banned as of May 28, the effective date of the new regulation. The remaining eight chloralkali plants that use asbestos diaphragms will be required to transition to alternative materials; six of these facilities were expected to complete this conversion within 5 years, and the other two were projected to follow at a later date. The EPA ordered most other uses of asbestos to cease from 6 months to 2 years after the effective date of the rule. Asbestos-containing sheet gaskets used to produce titanium dioxide and in the processing of nuclear material will have a 5-year phaseout, and the U.S. Department of Energy's Savannah River site will be permitted to use asbestos-containing sheet gaskets in the disposal of nuclear materials through 2037. In 2019, the EPA banned all discontinued

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uses of asbestos from restarting without the EPA having an opportunity to evaluate each intended use and take any necessary regulatory action. Consequently, the March 2024 rule will effectively prohibit all uses of asbestos in the United States as of the compliance date for each specific application.

Globally, asbestos is used predominantly in cement pipe, roofing sheets, and other construction materials in Asia. Worldwide consumption of unmanufactured asbestos fiber ranged from an estimated 1.1 million to 1.4 million tons per year from 2015 through 2024, a significant decrease from approximately 2 million tons in 2000, and will likely remain steady for the foreseeable future.

World Mine Production and Reserves: In addition to the countries listed, Zimbabwe may have produced asbestos from old mine tailings; information on the status of these operations was unavailable.

	Mine production		Reserves[7]
	2023	2024[e]	
United States	—	—	Small
Brazil	[8]189,000	160,000	11,000,000
China	[e]200,000	200,000	18,000,000
Kazakhstan	255,000	210,000	Large
Russia	600,000	600,000	110,000,000
World total (rounded)	1,240,000	1,200,000	Large

World Resources:[7] Reliable evaluations of global asbestos resources have not been published recently, and available information was insufficient to make accurate estimates for many countries. However, world resources are large and more than adequate to meet anticipated demand in the foreseeable future. Resources in the United States are composed mostly of short-fiber asbestos for which use in asbestos-based products is more limited than long-fiber asbestos.

Substitutes: Numerous materials substitute for asbestos, including calcium silicate, carbon fiber, cellulose fiber, ceramic fiber, glass fiber, steel fiber, wollastonite, and several organic fibers, such as aramid, polyethylene, polypropylene, and polytetrafluoroethylene. Several nonfibrous minerals or rocks, such as perlite, serpentine, silica, and talc, are also considered to be possible asbestos substitutes for products in which the reinforcement properties of fibers are not required. Membrane cells and mercury cells are alternatives to asbestos diaphragms used in the chloralkali industry.

[e]Estimated. NA Not available. — Zero.

[1]Source: U.S. Environmental Protection Agency, 2024, Asbestos part 1—Chrysotile asbestos—Regulation of certain conditions of use under the Toxic Substances Control Act (TSCA): Federal Register, v. 89, no. 61, March 28, p. 21970–22010. (Accessed October 15, 2024, at https://www.govinfo.gov/content/pkg/FR-2024-03-28/pdf/2024-05972.pdf.)

[2]Includes unmanufactured asbestos fiber (chrysotile) only; excludes asbestos contained in manufactured products.

[3]Modified from reported U.S. Census Bureau data. Additional chrysotile imports from China were reported in 2021 (59 tons) and 2022 (99 tons), but bill of lading information and data reported by the Government of China suggest that these shipments were misclassified. The U.S. Census Bureau also reported imports of 2 tons from Poland in 2023 and 4 tons from Germany through August 2024, but asbestos bans in these countries suggest that these shipments were misclassified.

[4]Exports of unmanufactured asbestos fiber were reported by the U.S. Census Bureau in each year from 2020 through 2024 but these shipments likely consisted of materials misclassified as asbestos, reexports, and (or) waste products because asbestos has not been mined in the United States since 2002.

[5]To account for year-to-year fluctuations in chrysotile imports owing to cycles of companies replenishing and drawing down stockpiles, consumption was estimated as a 5-year rolling average of imports for consumption. Information regarding the quantity of industry stocks was unavailable.

[6]Defined as imports – exports. The United States has been 100% import reliant since 2002. All domestic consumption of chrysotile was from imports and unreported inventories.

[7]See Appendix C for resource and reserve definitions and information concerning data sources.

[8]Export sales reported by the only producer of asbestos in Brazil. In February 2023, the Supreme Federal Court of Brazil confirmed a 2017 judgment that the extraction, sale, and use of asbestos were unconstitutional. Despite the ruling, a company in Brazil continued to mine asbestos, citing the authority of a State law that authorized extraction and processing for export purposes.

BARITE

(Data in thousand metric tons unless otherwise specified)

Domestic Production and Use: In 2024, three companies mined barite at four operations in Nevada. Mine production increased, but data were withheld to avoid disclosing company proprietary data. An estimated 2.3 million tons of barite (from domestic production and imports) was sold by crushers and grinders operating in nine States.

Typically, more than 90% of the barite sold in the United States is used as a weighting agent in fluids used in the drilling of oil and natural gas wells. The majority of Nevada crude barite was ground in Nevada and then sold to companies drilling in the Central and Western United States. Because of the higher cost of rail and truck transportation compared to ocean freight, offshore drilling operations in the Gulf of Mexico and onshore drilling operations in other regions primarily used imported barite.

Barite also is used as a filler, extender, or weighting agent in products such as paints, plastics, and rubber. Some specific applications include use in automobile brake and clutch pads, in automobile paint primer for metal protection and gloss, as a weighting agent in rubber, and in the cement jacket around underwater petroleum pipelines. In the metal-casting industry, barite is part of the mold-release compounds. Because barite significantly blocks X-ray and gamma-ray emissions, it is used as aggregate in high-density concrete for radiation shielding around X-ray units in hospitals, nuclear powerplants, and university nuclear research facilities. Ultrapure barite is used as a contrast medium in X-ray and computed tomography examinations of the gastrointestinal tract.

Salient Statistics—United States:	**2020**	**2021**	**2022**	**2023**	**2024[e]**
Production:					
Sold or used, mine	W	W	W	W	W
Ground and crushed[1]	1,410	1,670	2,220	2,260	2,300
Imports:[2]					
For consumption	1,480	1,660	2,330	2,420	2,000
General	869	1,440	1,890	2,220	1,900
Exports[3]	48	62	87	75	65
Consumption, apparent (crude and ground)[4]	W	W	W	W	W
Price, average unit value, ground, ex-works, dollars per metric ton	183	167	145	223	220
Employment, mine and mill, number[e]	360	330	380	440	440
Net import reliance[5] as a percentage of apparent consumption	>75	>75	>75	>75	>75

Recycling: None.

Import Sources (2020–23): India, 40%; China,[6] 25%; Morocco, 17%; Mexico, 14%; and other, 4%.

Tariff: Item	**Number**	**Normal Trade Relations 12–31–24**
Ground barite	2511.10.1000	Free.
Crude barite	2511.10.5000	$1.25 per metric ton.
Barium compounds:		
Barium oxide, hydroxide, and peroxide	2816.40.2000	2% ad valorem.
Barium chloride	2827.39.4500	4.2% ad valorem.
Barium sulfate, precipitated	2833.27.0000	0.6% ad valorem.
Barium carbonate, precipitated	2836.60.0000	2.3% ad valorem.

Depletion Allowance: 14% (domestic and foreign).

Government Stockpile: None.

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BARITE

Events, Trends, and Issues: Rig counts for oil and gas production are often used as an indicator of barite consumption. However, barite use per rig has been increasing owing to deeper oil and gas wells that require fewer rigs for oil and gas production. Through October 2024, the world annual average rig count excluding the United States was 1,169 compared with 1,154 in 2023. Increases in worldwide rig counts contributed to a slight increase in world barite production. At the beginning of 2024, the number of drill rigs[7] operating in the United States was 622; by the end of October 2024, the number of rigs operating had declined to 585. Rig counts remained 39% lower than that in the same period in 2019 before the global coronavirus disease 2019 (COVID-19) pandemic in 2020. Despite the slowing pace of domestic drill counts, barite sales were estimated to have increased slightly.

World Mine Production and Reserves: In response to concerns about dwindling global reserves of 4.2-specific-gravity barite used by the oil- and gas-drilling industry, the American Petroleum Institute issued an alternate specification for 4.1-specific-gravity weighting agents in 2010. Estimated reserves data were included only if developed since the adoption of the 4.1-specific-gravity standard. Reserves for China, Iran, and Turkey were revised based on company and Government reports.

	Mine production[e]		Reserves[8]
	2023	2024	
United States	W	W	NA
China	2,000	2,100	110,000
India	2,600	2,600	51,000
Iran	300	310	100,000
Kazakhstan	650	650	85,000
Laos	300	300	NA
Mexico	320	330	NA
Morocco	1,000	1,000	NA
Pakistan	130	130	NA
Russia	250	250	12,000
Turkey	[9]216	220	34,000
Other countries	314	320	NA
World total (rounded)	[10]8,080	[10]8,200	NA

World Resources:[8] In the United States, identified resources of barite were estimated to be 150 million tons, and undiscovered resources contributed an additional 150 million tons. The world's barite resources in all categories were about 2 billion tons, but only about 740 million tons were identified resources.

Substitutes: Owing to technical and economic factors, there are no large-scale alternatives to barite in oil- and gas-drilling fluids. Calcium carbonate, hematite, ilmenite, and manganese tetroxide are the most common alternatives used in specific circumstances. Some technical literature and patents also mention use of celestite, iron carbonate, and strontium carbonate, but these are not estimated to be widely used.

[e]Estimated. NA Not available. W Withheld to avoid disclosing company proprietary data.

[1]Imported and domestic barite, crushed and ground, sold or used by domestic grinding establishments.

[2]Includes data for the following Harmonized Tariff Schedule of the United States codes: 2511.10.1000, 2511.10.5000, and 2833.27.0000. General imports and imports for consumption data differ because of barite processed in free trade zones. General import data reports the form of imported barite at the time it entered the United States, whereas imports for consumption data reports crude barite processed in free trade zones as ground. Imports for consumption may not be immediately reported depending on processing time.

[3]Includes data for the following Schedule B numbers: 2511.10.1000 and 2833.27.0000.

[4]Defined as mine production (sold or used) + imports for consumption – exports.

[5]Defined as imports for consumption – exports.

[6]Includes Hong Kong.

[7]Source: Baker Hughes Co., 2024, North America rotary rig count: Baker Hughes Co. (Accessed October 24, 2024, at https://bakerhughesrigcount.gcs-web.com/na-rig-count.)

[8]See Appendix C for resource and reserve definitions and information concerning data sources.

[9]Reported.

[10]Excludes U.S. production.

BAUXITE AND ALUMINA[1]

(Data in thousand metric dry tons unless otherwise specified)

Domestic Production and Use: In 2024, a limited amount of bauxite and bauxitic clay was produced for nonmetallurgical use in Arkansas and Georgia. Production statistics were withheld for bauxite and estimated for alumina to avoid disclosing company proprietary data. In 2024, the reported quantity of bauxite consumed was estimated to be 1.8 million tons, 12% less than that reported in 2023, with an estimated value of about $54 million. About 76% of the bauxite consumed was refined by the Bayer process for alumina or aluminum hydroxide, and the remainder went to products such as abrasives, cement, chemicals, proppants, and refractories, and as a slag adjuster in steel mills. Alumina production was estimated to be 810,000 tons, 5% less than that in 2023. About 69% of the alumina produced went to primary aluminum smelters, and the remainder went to nonmetallurgical products, such as abrasives, ceramics, chemicals, and refractories.

Salient Statistics—United States:	**2020**	**2021**	**2022**	**2023**	**2024[e]**
Bauxite:					
Production, mine	W	W	W	W	W
Imports for consumption[2]	3,760	3,880	3,630	3,160	2,800
Exports[2]	16	13	10	14	13
Stocks, industry, yearend[e, 2]	250	200	200	240	250
Consumption:					
Apparent[3]	W	W	W	W	W
Reported	3,330	2,790	2,170	2,050	1,800
Price, average unit value of imports, free alongside ship (f.a.s.), dollars per metric ton	30	31	32	31	30
Net import reliance[4] as a percentage of apparent consumption	>75	>75	>75	>75	>75
Alumina:					
Production, refinery[e, 5]	1,300	1,000	920	850	810
Imports for consumption[5]	1,340	1,550	1,880	1,360	1,300
Exports[5]	153	180	174	139	150
Stocks, industry, yearend[5]	234	202	194	190	180
Consumption, apparent[3]	2,530	2,410	2,640	2,080	2,000
Price, average unit value of imports, f.a.s., dollars per metric ton	394	462	518	481	570
Net import reliance[4] as a percentage of apparent consumption	49	58	65	59	59

Recycling: None.

Import Sources (2020–23): Bauxite:[2] Jamaica, 67%; Turkey, 9%; Guyana, 8%; Australia, 6%; and other, 10%. Alumina:[5] Brazil, 68%; Jamaica, 10%; Australia, 7%; Canada, 5%; and other, 10%.

Tariff: Item	**Number**	**Normal Trade Relations 12–31–24**
Bauxite, calcined (refractory grade)	2606.00.0030	Free.
Bauxite, calcined (other)	2606.00.0060	Free.
Bauxite, crude dry (metallurgical grade)	2606.00.0090	Free.
Aluminum oxide (alumina)	2818.20.0000	Free.
Aluminum hydroxide	2818.30.0000	Free.

Depletion Allowance: 22% (domestic), 14% (foreign).

Government Stockpile: None.

Events, Trends, and Issues: In 2024, one domestic alumina refinery produced alumina from imported bauxite. A 1.2-million-ton-per-year alumina refinery in Gramercy, LA, produced alumina for aluminum smelting and specialty-grade alumina. A 500,000-ton-per-year alumina refinery in Burnside, LA, was temporarily shut down in August 2020 and remained idle in 2024. No plans were announced regarding its reopening. The average prices, f.a.s., for U.S. imports for consumption of crude dry bauxite and metallurgical-grade alumina during the first 8 months of 2024 were $30 per ton and $560 per ton, 5% less and 12% more than those in the same period in 2023, respectively.

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BAUXITE AND ALUMINA

A United States-based multinational aluminum producer acquired full control of an Australia-based joint-venture that mined bauxite and produced alumina and aluminum globally. An India-based aluminum producer increased production at its newly expanded alumina refinery in Odisha, to 3.5 million tons per year. In Australia, a 2.2-million-ton-per-year alumina refinery located in Western Australia was idled owing to market conditions and operating costs. A fire at a natural gas pipeline in Queensland, Australia, caused an alumina refinery to declare force majeure. An Indonesia-based mining company began production at its 2.2-million-ton-per-year alumina refinery in the Province of West Kalimantan. A 1.7-million-ton-per-year alumina refinery in Ukraine has remained closed since 2022 owing to the Russia-Ukraine conflict.

World Alumina Refinery and Bauxite Mine Production and Bauxite Reserves: Reserves for China, Indonesia, Kazakhstan, Turkey, and Vietnam were revised based on company and Government reports.

	Alumina production[5]		Bauxite production		Bauxite reserves[6]
	2023	2024[e]	2023	2024[e]	
United States	[e]850	720	W	W	20,000
Australia	18,800	18,000	104,000	100,000	[7]3,500,000
Brazil	[e]11,000	11,000	[e]32,000	33,000	2,700,000
Canada	1,500	1,500	—	—	—
China	82,400	84,000	[e]91,000	93,000	680,000
Germany	[e]900	960	—	—	—
Greece	869	860	[e]1,200	1,200	—
Guinea	273	300	123,000	130,000	7,400,000
India	[e]7,500	7,600	23,400	25,000	650,000
Indonesia	[e]1,200	1,300	[e]30,000	32,000	2,800,000
Ireland	1,380	1,600	—	—	—
Jamaica	1,400	1,500	6,000	6,100	2,000,000
Kazakhstan	1,300	1,400	4,560	4,900	280,000
Russia	3,020	2,900	[e]5,800	6,300	480,000
Saudi Arabia	1,830	1,800	[e]5,400	5,800	180,000
Spain	716	820	—	—	—
Turkey	300	320	2,940	3,200	69,000
United Arab Emirates	2,480	2,400	—	—	—
Vietnam	1,490	1,500	[e]3,920	4,200	3,100,000
Other countries	1,300	1,500	5,260	5,700	5,300,000
World total (rounded)	141,000	142,000	[8]438,000	[8]450,000	29,000,000

World Resources:[6] Bauxite resources are estimated to be between 55 billion and 75 billion tons, distributed in Africa (32%), Oceania (23%), South America and the Caribbean (21%), Asia (18%), and elsewhere (6%). Domestic resources of bauxite are inadequate to meet long-term U.S. demand, but the United States and most other major aluminum-producing countries have essentially inexhaustible subeconomic resources of aluminum in materials other than bauxite.

Substitutes: Bauxite is the only raw material used in the production of alumina on a commercial scale in the United States. Although currently not economically competitive with bauxite, vast resources of clay are technically feasible sources of alumina. Other raw materials, such as alunite, anorthosite, coal wastes, and oil shales, offer additional potential alumina sources. Synthetic mullite, produced from kaolin, bauxitic kaolin, kyanite, and sillimanite, substitutes for bauxite-based refractories. Silicon carbide and alumina zirconia can substitute for alumina and bauxite in abrasives but cost more.

[e]Estimated. W Withheld to avoid disclosing company proprietary data. — Zero.

[1]See also the Aluminum chapter. As a general rule, 4 tons of dried bauxite is required to produce 2 tons of alumina, which, in turn, can be used to produce 1 ton of aluminum.

[2]Includes all forms of bauxite, expressed as dry equivalent weights.

[3]Defined as production + imports – exports ± adjustments for industry stock changes.

[4]Defined as imports – exports ± adjustments for industry stock changes.

[5]Calcined equivalent weights.

[6]See Appendix C for resource and reserve definitions and information concerning data sources.

[7]For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 1.6 billion tons.

[8]Excludes U.S. production.

BERYLLIUM

(Data in metric tons, beryllium content, unless otherwise specified)

Domestic Production and Use: One company in Utah mined bertrandite ore and converted it, along with imported beryl, into beryllium hydroxide. Some of the beryllium hydroxide was shipped to the company's plant in Ohio, where it was converted into metal, oxide, and downstream beryllium-copper master alloy, and some was sold. Estimated beryllium apparent consumption in 2024 was 170 tons and was valued at about $260 million based on the most recent beryllium price estimate. Based on sales revenues, approximately 20% of beryllium products were used in industrial components, 19% in aerospace and defense applications, 11% in automotive electronics, 8% in telecommunications infrastructure, 6% each in consumer electronics and energy applications, 2% in semiconductor applications, and 28% in other applications. Beryllium alloy strip and bulk products, the most common forms of processed beryllium, were used in all application areas. Most unalloyed beryllium metal and beryllium composite products were used in defense and scientific applications. The U.S. Department of Defense supports the availability of domestic beryllium to meet critical defense needs. In 2010, under the Defense Production Act, Title III program, a public-private partnership with the leading U.S. beryllium producer reestablished domestic production of beryllium metal.

Salient Statistics—United States:	2020	2021	2022	2023	2024[e]
Production, mine shipments	165	175	175	185	180
Imports for consumption[1]	48	49	39	25	20
Exports[2]	25	30	61	68	30
Shipments from Government stockpile[3]	3	7	9	NA	NA
Consumption:					
Apparent[4]	196	196	187	144	170
Reported, ore	170	170	170	180	180
Price, annual average unit value, beryllium-copper master alloy,[5] dollars per kilogram of contained beryllium	620	680	660	1,400	1,500
Stocks, ore, industry, yearend	30	35	10	10	10
Net import reliance[6] as a percentage of apparent consumption	16	11	6	E	E

Recycling: Beryllium was recovered from new scrap generated during the manufacture of beryllium products and from old scrap. Detailed data on the quantities of beryllium recycled were not available but may account for as much as 20% to 25% of total beryllium consumption. The leading U.S. beryllium producer managed a recycling program for all its beryllium products, recovering approximately 40% of the beryllium content of the new and old beryllium alloy scrap.

Import Sources (2020–23):[1] Kazakhstan, 39%; Latvia, 23%; Japan, 17%; Canada, 6%; and other, 15%.

Tariff: Item	Number	Normal Trade Relations 12–31–24
Beryllium ores and concentrates	2617.90.0030	Free.
Beryllium oxide and hydroxide	2825.90.1000	3.7% ad valorem.
Beryllium-copper master alloy	7405.00.6030	Free.
Beryllium-copper plates, sheets, and strip:		
Thickness of 5 millimeters (mm) or more	7409.90.1030	3% ad valorem.
Thickness of less than 5 mm:		
Width of 500 mm or more	7409.90.5030	1.7% ad valorem.
Width of less than 500 mm	7409.90.9030	3% ad valorem.
Beryllium:		
Unwrought, including powders	8112.12.0000	8.5% ad valorem.
Waste and scrap	8112.13.0000	Free.
Other	8112.19.0000	5.5% ad valorem.

Depletion Allowance: 22% (domestic), 14% (foreign).

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BERYLLIUM

Government Stockpile:[7]

Material	FY 2024 Potential acquisitions	FY 2024 Potential disposals	FY 2025 Potential acquisitions	FY 2025 Potential disposals
Metal (all types)	—	7	—	7

Events, Trends, and Issues: Apparent consumption in 2024 increased by 18% from that in 2023 owing primarily to a 56% decrease in estimated beryllium exports offset by a 20% decrease in estimated imports. The decrease in exports reflected a large reduction in beryllium metal exports to Canada, Germany, Japan, and Mexico. The decrease in imports reflected a reduction in beryllium metal imports from Belgium, Kazakhstan, and Latvia. During the first 6 months of 2024, the leading U.S. beryllium producer reported that net sales of its beryllium alloy strip and bulk products and beryllium metal and composite products were 4% lower than those during the first 6 months of 2023. Net sales of beryllium products decreased primarily in the automotive, industrial components, and telecommunications and data markets. Because of the toxic nature of beryllium, various international, national, and State guidelines and regulations have been established regarding beryllium in air, water, and other media. Industry is required to carefully control the quantity of beryllium dust, fumes, and mists in the workplace.

World Mine Production and Reserves:

	Mine production[8, 9] 2023	Mine production[8, 9] 2024[e]	Reserves[10]
United States	185	180	The United States has very little beryl that can be economically hand sorted from pegmatite deposits. An epithermal deposit in the Spor Mountain area in Utah is a large bertrandite resource, which is being mined. Proven and probable bertrandite reserves in Utah total about 19,000 tons of beryllium content. World beryllium reserves were not available.
Brazil	[e]40	80	
China	[e]74	77	
Madagascar	[e]1	1	
Mozambique	23	24	
Rwanda	[e]1	1	
Uganda	[e]1	1	
World total (rounded)	325	360	

World Resources:[10] The world's identified resources of beryllium have been estimated to be more than 100,000 tons. About 60% of these resources are in the United States; by tonnage, the Spor Mountain area in Utah, the McCullough Butte area in Nevada, the Black Hills area in South Dakota, the Sierra Blanca area in Texas, the Seward Peninsula in Alaska, and the Gold Hill area in Utah account for most of the total.

Substitutes: Because the cost of beryllium is high compared with that of other materials, it is used in applications in which its properties are crucial. In some applications, certain metal matrix or organic composites, high-strength grades of aluminum, pyrolytic graphite, silicon carbide, steel, or titanium may be substituted for beryllium metal or beryllium composites. Copper alloys containing nickel and silicon, tin, titanium, or other alloying elements or phosphor bronze alloys (copper-tin-phosphorus) may be substituted for beryllium-copper alloys, but these substitutions can result in substantially reduced performance. Aluminum nitride or boron nitride may be substituted for beryllium oxide.

[e]Estimated. E Net exporter. NA Not available. — Zero.

[1]Includes estimated beryllium content of imported ores and concentrates, oxide and hydroxide, unwrought metal (including powders), beryllium articles, waste and scrap, beryllium-copper master alloy, and beryllium-copper plates, sheets, and strip.

[2]Includes estimated beryllium content of exported unwrought metal (including powders), beryllium articles, and waste and scrap.

[3]Change in total inventory from prior yearend inventory. If negative, increase in inventory. Beginning in 2023, Government stock changes no longer available.

[4]Defined for 2020–22 as production + imports − exports ± adjustments for Government and industry stock changes. Beginning in 2023, Government stock changes no longer included.

[5]Calculated from gross weight and customs value of imports; beryllium content estimated to be 4%. Rounded to two significant figures.

[6]Defined for 2020–22 as imports − exports ± adjustments for Government and industry stock changes. Beginning in 2023, Government stock changes no longer included.

[7]See Appendix B for definitions.

[8]In addition to the countries listed, Kazakhstan and Portugal may have produced beryl ore, but available information was inadequate to make reliable estimates of output. Other nations that produced gemstone beryl ore may also have produced some industrial beryl ore.

[9]Based on 4% beryllium content of bertrandite and beryl sources.

[10]See Appendix C for resource and reserve definitions and information concerning data sources.

BISMUTH

(Data in metric tons unless otherwise specified)

Domestic Production and Use: The United States ceased production of primary refined bismuth in 1997 and is highly import reliant. Bismuth is contained in some lead ores mined domestically. However, the last domestic primary lead smelter closed at yearend 2013; since then, all lead concentrates have been exported for smelting.

Most domestic bismuth consumption was for chemicals used in cosmetic, industrial, laboratory, and pharmaceutical applications. Bismuth use in pharmaceuticals included bismuth subsalicylate (the active ingredient in over-the-counter stomach remedies) and other compounds used to treat burns, intestinal disorders, and stomach ulcers. Bismuth compounds such as bismuth nitrate, bismuth oxychloride, and bismuth vanadate are also used in industrial applications for the manufacture of ceramic glazes, crystalware, high-performance pigments, and pearlescent pigments.

Bismuth has a wide variety of metallurgical applications, including use as an additive to improve metal integrity of malleable cast iron in the foundry industry and as a nontoxic replacement for lead in brass, free-machining aluminum alloys and steels, and solders. The use of bismuth in brass for pipe fittings, fixtures, and water meters increased after 2014, when the definition of “lead-free” under the Safe Drinking Water Act was modified to reduce the maximum lead content of “lead-free” pipes and plumbing fixtures to 0.25% from 8%. The melting point of bismuth is relatively low at 271 degrees Celsius. Bismuth is an important component of various fusible alloys that can be used in holding devices for grinding optical lenses, as plugs for abandoned oil wells, as a temporary filler to prevent damage to tubes in bending operations, as a triggering mechanism for fire sprinklers, and in other applications in which a low melting point is ideal. Bismuth-tellurium-oxide alloy film paste is used in the manufacture of semiconductor devices.

Salient Statistics—United States:	**2020**	**2021**	**2022**	**2023**	**2024[e]**
Production:					
Refinery	—	—	—	—	—
Secondary (scrap)[e]	80	80	80	80	80
Imports for consumption, metal, alloys, and scrap:					
Containing more than 99.99% bismuth, by weight	NA	NA	740	731	650
Other	NA	NA	2,340	1,110	1,200
Total[1]	1,650	1,980	3,080	1,840	1,800
Exports, metal, alloys, and scrap:					
Containing more than 99.99% bismuth, by weight	NA	NA	144	131	560
Other	NA	NA	360	329	600
Total[2]	699	1,010	503	460	1,200
Consumption:					
Apparent[3]	1,210	1,030	2,600	1,450	760
Reported	513	597	724	691	700
Price, average,[4] dollars per pound	2.73	3.74	3.90	4.08	5.30
Stocks, yearend, consumer, bismuth metal	271	297	356	365	365
Net import reliance[5] as a percentage of apparent consumption	93	92	97	94	89

Recycling: Recycled bismuth-containing alloy scrap was thought to compose up to 3% to 10% of U.S. bismuth apparent consumption for the years 2020–24.

Import Sources (2020–23): China,[6] 67%; Republic of Korea, 23%; and other, 10%.

Tariff: Item	**Number**	**Normal Trade Relations 12–31–24**
Bismuth and articles thereof, including waste and scrap:		
Containing more than 99.99% of bismuth, by weight	8106.10.0000	Free.
Other	8106.90.0000	Free.

Depletion Allowance: 22% (domestic), 14% (foreign).

Government Stockpile: None.

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BISMUTH

Events, Trends, and Issues: In 2024, average monthly prices for bismuth (in-warehouse, Rotterdam) increased from $3.89 per pound in January to $6.29 per pound in October. The estimated annual average price in 2024 was $5.30 per pound, a 30% increase from that in 2023, and the highest annual average price since 2018. Bismuth metal prices have been increasing worldwide and particularly in China since 2023. China, the leading producer and exporter of bismuth, reportedly experienced high feedstock prices as competition for bismuth ore increased among domestic smelters. United States bismuth metal imports (under Harmonized System code 8106) from China decreased by 40% to 580 tons in 2024 from 964 tons in 2023.

Estimated world production of bismuth was 16,000 tons in 2024 compared with 16,200 tons in 2023. Reported bismuth production capacities were unavailable.

World Refinery Production and Capacity:

	Refinery production[e]		Production capacity
	2023	2024[e]	
United States	—	—	NA
Bolivia	68	70	NA
Bulgaria	46	50	NA
China	13,300	13,000	NA
Japan	500	500	NA
Kazakhstan	180	180	NA
Korea, Republic of	1,000	1,000	NA
Laos	[8]1,150	1,100	NA
World total (rounded)	16,200	16,000	NA

World Resources:[7] Bismuth reserves and resources data were generally not reported at a mine or country level and thus difficult to quantify. Bismuth minerals rarely occur in sufficient quantities to be mined as principal products; bismuth is produced most often as a byproduct during the processing of lead ores. In China and Vietnam, bismuth is also produced as a byproduct or coproduct of tungsten and other metal ore processing. In Japan and the Republic of Korea, bismuth is produced as a byproduct or coproduct of zinc ore processing. The Tasna Mine in Bolivia, which has been inactive since 1996, and a mine in China are the only mines where bismuth has been the primary product.

Substitutes: Bismuth compounds can be replaced in pharmaceutical applications by alumina, antibiotics, calcium carbonate, and magnesia. Titanium-dioxide-coated mica flakes and fish-scale extracts are substitutes in certain pigment uses. Cadmium, indium, lead, and tin can partially replace bismuth in low-temperature solders. Resins can replace bismuth alloys for holding metal shapes during machining, and glycerin-filled glass bulbs can replace bismuth alloys in triggering devices for fire sprinklers. Free-machining alloys can contain lead, selenium, or tellurium as a replacement for bismuth. Bismuth is an environmentally friendly substitute for lead in plumbing and many other applications, including fishing weights, hunting ammunition, lubricating greases, and soldering alloys.

[e]Estimated. NA Not available. — Zero.

[1]Includes data for the following Harmonized Tariff Schedule of the United States codes: 8106.00.0000 (for the years 2020–21), and 8106.10.0000 and 8106.90.0000 (for the years 2022–24).

[2]Includes data for the following Schedule B numbers: 8106.00.0000 (for the years 2020–21), and 8106.10.0000 and 8106.90.0000 (for the years 2022–24).

[3]Defined as secondary production + imports – exports ± adjustments for industry stock changes.

[4]Prices are based on 99.99%-purity metal at warehouse (Rotterdam) in minimum lots of 1 ton. Source: Fastmarkets.

[5]Defined as imports – exports ± adjustments for industry stock changes.

[6]Includes Hong Kong.

[7]See Appendix C for resource and reserve definitions and information concerning data sources.

[8]Reported.

BORON

(Data in thousand metric tons unless otherwise specified)

Domestic Production and Use: Three companies in southern California produced borates in 2024, and most of the boron products consumed in the United States were manufactured domestically. Estimated boron production increased in 2024 compared with production in 2023. U.S. boron production and consumption data were withheld to avoid disclosing company proprietary data. The leading boron producer mined borate ores, which contain the minerals kernite, tincal, and ulexite, by open pit methods and operated associated compound plants. Kernite was used to produce boric acid, tincal was used to produce sodium borate, and ulexite was used as a primary ingredient in the manufacture of a variety of specialty glasses and ceramics. A second company produced borates from brines extracted through solution-mining techniques. A third company began mining borates using solution mining techniques in January 2024. Boron minerals and chemicals were principally consumed in the north-central and eastern United States. In 2024, the glass and ceramics industries remained the leading domestic users of boron products. Boron also was used as a component in abrasives, cleaning products, insecticides, insulation, and in the production of semiconductors.

Salient Statistics—United States:	**2020**	**2021**	**2022**	**2023**	**2024[e]**
Production	W	W	W	W	W
Imports for consumption:					
Refined borax	174	232	168	156	160
Boric acid	39	54	48	38	42
Colemanite (calcium borates)	18	3	1	2	1
Ulexite (sodium borates)	41	49	38	20	37
Exports:					
Boric acid	257	280	239	253	240
Refined borax	594	607	651	604	590
Consumption, apparent[1]	W	W	W	W	W
Price, average unit value of imports, cost, insurance, and freight, dollars per metric ton	380	394	485	606	560
Employment, number	1,330	1,330	1,400	1,430	1,500
Net import reliance[2] as a percentage of apparent consumption	E	E	E	E	E

Recycling: Insignificant.

Import Sources (2020–23): All forms: Turkey, 90%; Bolivia, 6%; and other, 4%.

Tariff: Item	**Number**	**Normal Trade Relations 12–31–24**
Natural borates:		
Sodium (ulexite)	2528.00.0005	Free.
Calcium (colemanite)	2528.00.0010	Free.
Boric acids	2810.00.0000	1.5% ad valorem.
Borates, refined borax:		
Anhydrous	2840.11.0000	0.3% ad valorem.
Non-anhydrous	2840.19.0000	0.1% ad valorem.

Depletion Allowance: Borax, 14% (domestic and foreign).

Government Stockpile: None.

Events, Trends, and Issues: Elemental boron is a metalloid with limited commercial applications. Although the term “boron” is commonly referenced, it does not occur in nature in an elemental state. Boron combines with oxygen and other elements to form boric acid or inorganic salts called borates. Boron compounds, chiefly borates, are commercially important; therefore, boron products are priced and sold based on their boric oxide (B_2O_3) content, varying by ore and compound and by the absence or presence of calcium and sodium. Four borate minerals—colemanite, kernite, tincal, and ulexite—account for 90% of the borate minerals used by industry worldwide. Although borates were used in more than 300 applications, more than three-quarters of world consumption was used in ceramics, detergents, fertilizers, and glass.

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BORON

China, India, Canada, Indonesia, and Mexico, in decreasing order of tonnage, were the countries that imported the largest quantities of refined borates from the United States in 2024. Domestic shipments of boric acid were sent to China, the Netherlands, the Republic of Korea, Taiwan, and Brazil, in decreasing order of tonnage. Because China has low-grade boron reserves and demand for boron is anticipated to rise in that country, imports from the United States were expected to remain steady during the next several years.

Interests and investments in boron derivatives continued domestically and abroad. The U.S. Department of Defense under the Defense Production Act, Title III, awarded $49.6 million to a company headquartered in Golden, CO, in December 2023. The funding was to be used to increase domestic boron carbide production capacity. In April 2024, a domestic company began boric acid production at its small-scale boron facility in Newberry Springs, CA. It began mining borates in January 2024. This facility's initial production capacity was about 1,800 tons per year, and the company planned to increase production to about 8,200 tons per year in the future. The small-scale boron facility was expected to focus on specialty boron products for industries related to defense, electric transportation, food security, and global decarbonization.

One Australia-based mine developer progressed toward construction of its boric acid project in Nevada. In September 2024, the Bureau of Land Management completed its final environmental impact statement and determined that development of the project may proceed. Once constructed, the project was expected to have a 26-year mine life and produce about 175,000 tons per year of boric acid. Initial production was expected to begin in 2028.

About 18 months after opening its first boron carbide facility in March 2023, Turkey opened another boron facility, the Bigadiç Granular Boron Production Facility, in September 2024. The new facility was expected to primarily produce granulated pipes for the fertilizer industry. This facility has a production capacity of 35,000 tons per year. By the end of August 2024, borate production in Turkey had increased by 36% compared with that in the same period in 2023.

World Production and Reserves: Reserve data for China were revised based on Government reports.

	Production—All forms[e]		Reserves[3]
	2023	2024	
United States	W	W	48,000
Argentina, crude ore	160	160	NA
Bolivia, ulexite	140	230	NA
Chile, ulexite	420	420	35,000
China, boric oxide equivalent	300	340	9,100
Germany, compounds	38	40	NA
Peru, crude borates	300	300	4,000
Russia, datolite ore	80	80	40,000
Turkey, refined borates	2,500	3,000	950,000
World total[4]	XX	XX	XX

World Resources:[3] Deposits of borates are associated with volcanic activity and arid climates, with the largest economically viable deposits in the Mojave Desert of the United States, the Alpide belt along the southern margin of Eurasia, and the Andean belt of South America. U.S. deposits consist primarily of tincal, kernite, and borates contained in brines, and to a lesser extent, ulexite and colemanite. About 70% of all deposits in Turkey are colemanite, primarily used in the production of heat-resistant glass. At current levels of consumption, world resources are adequate for the foreseeable future.

Substitutes: The substitution of other materials for boron is possible in detergents, enamels, insulation, and soaps. Sodium percarbonate can replace borates in detergents and requires lower temperatures to undergo hydrolysis, which is an environmental consideration. Some enamels can use other glass-producing substances, such as phosphates. Insulation substitutes include cellulose, foams, and mineral wools. In soaps, sodium and potassium salts of fatty acids can act as cleaning and emulsifying agents.

[e]Estimated. E Net exporter. NA Not available. W Withheld to avoid disclosing company proprietary data. XX Not applicable.

[1]Defined as production + imports – exports.

[2]Defined as imports – exports.

[3]See Appendix C for resource and reserve definitions and information concerning data sources.

[4]World totals cannot be calculated because production and reserves are not reported in a consistent manner by all countries.

BROMINE

(Data in metric tons, bromine content, unless otherwise specified)

Domestic Production and Use: Bromine was recovered from underground brines by two companies in Arkansas. Bromine is one of the leading mineral commodities, in terms of value, produced in Arkansas. The two bromine companies in the United States account for a large percentage of world production capacity.

The leading global applications of bromine are for the production of brominated flame retardants (BFRs) and clear brine drilling fluids. Bromine compounds also are used in a variety of other applications, including industrial uses, as intermediates, and for water treatment. U.S. apparent consumption of bromine in 2024 was estimated to be more than that in 2023.

Salient Statistics—United States:	2020	2021	2022	2023	2024[e]
Production	W	W	W	W	W
Imports for consumption, elemental bromine and compounds[1]	30,700	27,200	36,500	50,800	61,000
Exports, elemental bromine and compounds[2]	36,600	27,900	19,400	38,900	34,000
Consumption, apparent[3]	W	W	W	W	W
Price, average unit value of imports (cost, insurance, and freight), dollars per kilogram, bromine content	2.67	2.85	3.29	2.92	2.70
Employment, number[e]	1,100	1,100	1,100	1,100	1,100
Net import reliance[4] as a percentage of apparent consumption	E	E	<25	<25	<25

Recycling: Some bromide solutions were recycled to obtain elemental bromine and to prevent the solutions from being disposed of as hazardous waste. For example, hydrogen bromide is emitted as a byproduct of several organic reactions; this byproduct can be recycled with virgin bromine brines and used as a source of bromine production. Bromine contained in plastics, such as BFRs, can be difficult and costly to remove because the BFR is often bound to the polymer or resin matrix; therefore, bromine will often be recycled via the parent polymer with the polymer used again in new products. The stability of BFRs may reduce or eliminate the need for incorporating additional flame retardants into new products made from recycled plastic because the recycled plastic may meet the same levels of fire safety as the virgin material. However, this stability may lead to the unintentional reintroduction of bromine or BFRs into new plastic product cycles. Bromine used in zinc-bromine batteries can be removed and completely recovered as bromine at the battery's end of life, purified, and used for new batteries. Available information was insufficient to estimate the quantity of bromine recovered and recycled.

Import Sources (2020–23):[5] Israel, 83%; Jordan, 9%; China,[6] 3%; and other, 5%.

Tariff: Item	Number	Normal Trade Relations 12–31–23
Bromine	2801.30.2000	5.5% ad valorem.
Hydrobromic acid	2811.19.3000	Free.
Potassium or sodium bromide	2827.51.0000	Free.
Ammonium, calcium, or zinc bromide	2827.59.2500	Free.
Potassium bromate	2829.90.0500	Free.
Sodium bromate	2829.90.2500	Free.
Methyl bromide[7]	2903.61.0000	Free.
Ethylene dibromide[8]	2903.62.1000	5.4% ad valorem.
Dibromoneopentylglycol	2905.59.3000	Free.
Tetrabromobisphenol A	2908.19.2500	5.5% ad valorem.
Decabromodiphenyl and octabromodiphenyl oxide	2909.30.0700	5.5% ad valorem.

Depletion Allowance: Brine wells, 5% (domestic and foreign).

Government Stockpile: None.

Events, Trends, and Issues: The United States maintained its position as one of the leading bromine producers in the world along with China, Israel, and Jordan. In 2024, estimated total imports of bromine and bromine compounds (bromine content) increased by 20% from those in 2023, and the leading source of imports of bromine and bromide compounds (gross weight) through July 2024 was Israel (87%), followed by Jordan (10%). The average annual unit value of imported bromine and bromine compounds (bromine content) was approximately $2.70 per kilogram, which was 8% less than that in 2023. Together, the leading imported bromine products in terms of both gross weight and bromine content were bromides and bromide oxides of ammonium, calcium, or zinc and bromides of sodium or potassium, accounting for more than 90% of total imported bromine.

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BROMINE

In 2024, estimated total exports (bromine content) decreased by 13% compared with those in 2023, and the leading destinations for exports (gross weight) through July 2024 were Guyana (29%), Saudi Arabia (25%), and the United Kingdom (14%). The average annual unit value of exported bromine and bromine compounds (bromine content) was approximately $3.30 per kilogram, compared with $3.28 per kilogram in 2023.

In July 2024, the U.S. Food and Drug Administration (FDA) revoked the authorization for brominated vegetable oil (BVO) to be used as a food additive. BVO had been used to stabilize fruit flavoring oils in fruit-flavored beverages since the 1920s. It was classified as generally recognized as safe (GRAS) by the FDA until 1970 when it lost its GRAS classification owing to toxicity concerns. Based on available information at the time, the FDA approved its continued use at a maximum level of 15 parts per million in beverages compared with the previous maximum level of 150 parts per million. A more recent collaborative study demonstrated that a safe level of dietary BVO could not be established; therefore, there was no longer a reasonable certainty of no harm from the use of BVO as a stabilizer in flavoring oils.[9]

World Production and Reserves: Reserves for Jordan were revised based on company reports.

	Production[e]		Reserves[10]
	2023	2024	
United States	W	W	11,000,000
China	[11]101,000	100,000	130,000
India	6,900	7,000	NA
Israel	[11]143,000	140,000	Large
Japan	20,000	20,000	NA
Jordan	[11]116,000	120,000	360,000
Ukraine	8,000	8,000	NA
World total (rounded)	[12]395,000	[12]400,000	Large

World Resources:[10] Bromine is found principally in seawater, evaporitic (salt) lakes, and underground brines associated with petroleum deposits. Seawater contains about 65 parts per million bromine, or an estimated 100 trillion tons. The Dead Sea, in the Middle East, is estimated to contain 1 billion tons of bromine. Bromine also is recovered from seawater as a coproduct during evaporation to produce salt.

Substitutes: Chlorine and iodine may be substituted for bromine in a few chemical reactions and for sanitation purposes. There are no comparable substitutes for bromine in various oil- and gas-well-completion and packer applications. Because plastics have a low ignition temperature, aluminum hydroxide, magnesium hydroxide, organic chlorine compounds, and phosphorus compounds can be substituted for bromine as fire retardants in some uses.

[e]Estimated. E Net exporter. NA Not available. W Withheld to avoid disclosing company proprietary data.

[1]Includes data for the Harmonized Tariff Schedule of the United States codes shown in the "Tariff" section.

[2]Includes data for the following Schedule B numbers: 2801.30.2000, 2827.51.0000, and 2827.59.0000 (for the years 2020–24); 2903.31.0000 and 2903.39.1520 (for the years 2020–21); and 2903.61.0000 and 2903.62.1000 (for the years 2022–24).

[3]Defined as production (sold or used) + imports – exports.

[4]Defined as imports – exports.

[5]Calculated using the gross weight of imports.

[6]Includes Hong Kong.

[7]Prior to 2022, was listed under Harmonized Tariff Schedule of the United States code 2903.39.1520.

[8]Prior to 2022, was listed under Harmonized Tariff Schedule of the United States code 2903.31.0000.

[9]Source: U.S. Food and Drug Administration, 2024, Revocation of authorization for use of brominated vegetable oil in food: Federal Register, v. 89, no. 128, July 3, p. 55040–55045. (Accessed September 25, 2024, at https://www.govinfo.gov/content/pkg/FR-2024-07-03/pdf/2024-14300.pdf.)

[10]See Appendix C for resource and reserve definitions and information concerning data sources.

[11]Reported.

[12]Excludes U.S. production.

CADMIUM

(Data in metric tons unless otherwise specified)

Domestic Production and Use: One company operating in Tennessee recovered an estimated 300 tons of primary cadmium metal valued at $1.2 million as a byproduct of zinc leaching from roasted sulfide concentrates at the only domestic zinc smelter. In 2024, with a shift in focus to lithium-ion battery recycling, a company in Ohio that had been recovering secondary cadmium metal shut down its nickel cadmium (NiCd) battery recycling line. Another battery recycling company, established in Ohio in 2022, processed both consumer and industrial NiCd batteries for salable metals and was recovering cadmium metal in 2024 and planned to refine it to battery-grade purity. Cadmium metal and compounds are mainly consumed for NiCd batteries, but also for alloys, coatings, and pigments. An increasing use for cadmium was in cadmium-telluride (CdTe) thin-film solar panels, and in cadmium-zinc-telluride (CdZnTe) substrates for radiation detectors and imaging applications.

Salient Statistics—United States:	2020	2021	2022	2023	2024[e]
Production:					
Primary, refined[1]	211	241	212	375	300
Secondary	W	W	W	W	W
Imports for consumption:					
Unwrought cadmium and powders	282	155	99	72	7
Wrought cadmium and other articles	3	2	1	1	3
Cadmium waste and scrap	90	85	40	([2])	26
Cadmium oxide	28	14	33	37	17
Cadmium sulfide	4	—	([2])	—	20
Cadmium pigments and preparations based on cadmium compounds	69	101	146	147	120
Exports:					
Unwrought cadmium and powders	4	51	68	100	20
Wrought cadmium and other articles	482	217	60	21	30
Cadmium waste and scrap	([2])	—	2	14	—
Cadmium pigments and preparations based on cadmium compounds	2,120	550	747	947	500
Consumption of metal, apparent[3]	W	W	W	W	W
Price, metal, annual average,[4] dollars per kilogram	2.29	2.56	3.42	4.06	4.1
Net import reliance[5] as a percentage of apparent consumption	<75	<50	<25	E	E

Recycling: Secondary cadmium is mainly recovered from spent consumer and industrial NiCd batteries. Other waste and scrap from which cadmium can be recycled includes copper-cadmium alloy scrap, some complex nonferrous alloy scrap, cadmium-containing dust from electric-arc furnaces, and CdTe solar panels.

Import Sources (2020–23):[6] China,[7] 34%; Germany, 31%; Australia, 23%; Peru, 10%; and other, 2%.

Tariff: Item	Number	Normal Trade Relations 12–31–24
Cadmium oxide	2825.90.7500	Free.
Cadmium sulfide	2830.90.2000	3.1% ad valorem.
Pigments and preparations based on cadmium compounds	3206.49.6010	3.1% ad valorem.
Cadmium waste and scrap	8112.61.0000	Free.
Unwrought cadmium and powders	8112.69.1000	Free.
Wrought cadmium and other articles	8112.69.9000	4.4% ad valorem.

Depletion Allowance: 22% (domestic), 14% (foreign).

Government Stockpile:[8] The fiscal year (FY) 2025 potential acquisitions include 2,800 square centimeters of CdZnTe substrates, a 180% increase from 1,000 square centimeters in FY 2024.

Events, Trends, and Issues:

For the second consecutive year, the United States was a net exporter of cadmium. Average prices for cadmium decreased midyear, reflecting the seasonal buying patterns in India, and ended the year higher than those in January. India was the leading importer of cadmium metal; more than 11,400 tons were imported in 2023, and almost 6,000 tons as of August 2024. Though significant quantities of cadmium sponge were produced in India as a

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byproduct of zinc smelting, no production of cadmium metal was reported as of August 2024 in monthly statistics published by the Indian Bureau of Mines. Based on estimated production of cadmium as well as imports and exports, China replaced India as the leading consumer of cadmium in 2024.

Cadmium use in semiconductors continued to increase, especially CdTe in thin-film solar panels. The Inflation Reduction Act of 2022 provided tax incentives to transition to clean energy including for the domestic manufacturing of solar modules and components. The leading domestic CdTe solar panel manufacturer, based in Ohio, began commercial production in the third quarter of 2024 at a fourth facility, located in Alabama, that increased manufacturing capacity to almost 11 gigawatts (GW) per year. A fifth site was under construction in Louisiana and was expected to add another 3.5 GW per year in the second half of 2025. Worldwide, capacity was about 21 GW per year including a facility in India that opened in early 2024. In January, the National Renewable Energy Laboratory, administrator of the 3-year Cadmium Telluride Photovoltaics Accelerator Program, announced that $1.8 million had been awarded in a second round of contracts to support development of CdTe solar cells that would be more efficient and have a lower cost.

World Refinery Production and Reserves:

	Refinery production[e]		Reserves[9]
	2023	2024	
United States[1]	[10]375	300	Quantitative estimates of reserves were not available. The cadmium content of typical zinc ores averages about 0.03%. See the Zinc chapter for zinc reserves.
Australia	900	900	
Bulgaria	310	300	
Canada	1,800	1,700	
China	9,300	9,300	
Germany	—	170	
Japan	1,800	1,700	
Kazakhstan	1,000	1,000	
Korea, Republic of	4,500	4,500	
Mexico	[10]1,020	1,200	
Netherlands	[10]726	400	
Norway	420	350	
Peru	[10]494	620	
Poland	250	250	
Russia	1,000	800	
Uzbekistan	220	200	
World total (rounded)	24,100	24,000	

World Resources:[9] Cadmium is generally recovered from zinc ores and concentrates. Sphalerite, the most economically significant zinc ore mineral, commonly contains minor amounts of cadmium, which shares certain similar chemical properties with zinc and often substitutes for zinc in the sphalerite crystal lattice. The cadmium mineral greenockite is frequently associated with weathered sphalerite and wurtzite.

Substitutes: Batteries with other chemistries, particularly lithium-ion, can replace NiCd batteries in many applications. Except where the surface characteristics of a coating are critical (for example, fasteners for aircraft), coatings such as zinc-nickel can be substituted for cadmium in many plating applications. Cerium sulfide is used as a replacement for cadmium pigments, mostly in plastics. Barium stabilizers can replace barium-cadmium stabilizers in flexible polyvinyl chloride (PVC) applications. Thin-film technologies based on copper-indium-gallium-selenide and perovskite materials continued to be investigated but were not yet commercially feasible.

[e]Estimated. E Net exporter. W Withheld to avoid disclosing company proprietary data. — Zero.

[1]Cadmium metal produced as a byproduct of zinc refining.

[2]Less than ½ unit.

[3]Defined as primary production + secondary production + imports of unwrought cadmium and powders – exports of unwrought cadmium and powders.

[4]Average free market price for 99.95% purity in 10-ton lots; cost, insurance, and freight; global ports. Source: Fastmarkets MB.

[5]Defined as imports of unwrought cadmium and powders – exports of unwrought cadmium and powders.

[6]Unwrought cadmium and powders; Harmonized Tariff Schedule of the United States code 8107.20.0000 for 2019–21 and 8112.69.1000 beginning in 2022.

[7]Includes Hong Kong.

[8]See Appendix B for definitions.

[9]See Appendix C for resource and reserve definitions and information concerning data sources.

[10]Reported.

CEMENT

(Data in thousand metric tons unless otherwise specified)

Domestic Production and Use: In 2024, U.S. portland and blended cement production decreased by 4% to an estimated 84 million tons, and masonry cement production also decreased by 4% to an estimated 2.2 million tons. Cement was produced at 99 plants in 34 States and in Puerto Rico. Texas, Missouri, California, and Florida were, in descending order of production, the four leading cement-producing States and accounted for approximately 43% of U.S. production. Overall, the U.S. cement industry's growth continued to be constrained by closed or idle plants, underutilized capacity at others, production disruptions from plant upgrades, and relatively inexpensive imports. In 2024, shipments of cement were an estimated 110 million tons with an estimated value of $17 billion. In 2024, an estimated 70% to 75% of sales were to ready-mixed concrete producers, 12% to concrete product manufacturers, 8% to 10% to contractors, and 5% to 10% to other customer types.

Salient Statistics—United States:[1]	2020	2021	2022	2023	2024[e]
Production:					
Portland and masonry cement[2]	89,300	91,000	91,200	[e]90,000	86,000
Clinker	78,951	79,616	79,489	[e]77,000	73,000
Shipments to final customers, includes exports	104,580	108,969	111,092	[e]110,000	110,000
Imports for consumption:					
Hydraulic cement	15,531	19,937	24,985	24,986	24,000
Clinker	1,204	1,563	1,021	921	820
Exports, hydraulic cement and clinker	884	939	904	889	900
Consumption, apparent[3]	105,000	111,000	114,000	[e]110,000	110,000
Price, average mill unit value, dollars per metric ton	125	127	139	[e]150	160
Stocks, cement, yearend	7,180	6,280	8,020	[e]8,500	7,500
Employment, mine and mill, number[e]	12,200	12,300	12,800	13,000	13,000
Net import reliance[4] as a percentage of apparent consumption	15	19	22	22	22

Recycling: Cement is not recycled, but significant quantities of concrete are recycled for use as a construction aggregate. Cement kilns can use waste fuels, recycled cement kiln dust, and recycled raw materials such as slags and fly ash. Various secondary materials can be incorporated as supplementary cementitious materials (SCMs) in blended cements and in the cement paste in concrete.

Import Sources (2020–23):[5] Turkey, 32%; Canada, 22%; Vietnam, 10%; Greece, 9%; and other, 27%.

Tariff: Item	Number	Normal Trade Relations 12–31–24
Cement clinker	2523.10.0000	Free.
White portland cement	2523.21.0000	Free.
Other portland cement	2523.29.0000	Free.
Aluminous cement	2523.30.0000	Free.
Other hydraulic cement	2523.90.0000	Free.

Depletion Allowance: Not applicable. Certain raw materials for cement production have depletion allowances.

Government Stockpile: None.

Events, Trends, and Issues: The value of total construction put in place in the United States increased by 7% during the first 9 months of 2024 compared with that in the same period in 2023. Both residential and nonresidential construction spending increased; however, new privately owned housing units started through September 2024 decreased by 3% compared with those during the same period in 2023. Reported cement shipments decreased by 6% during the first 8 months of 2024 compared with those in the same period in 2023. The leading cement-consuming States continued to be Texas, Florida, and California, in descending order by tonnage.

According to the Bureau of Economic Analysis, the real gross domestic product (GDP) increased by 5% during the first 9 months of 2024 compared with the real GDP for full year 2023. The Federal Reserve lowered interest rates in 2024, and funding from the Bipartisan Infrastructure Law and the Inflation Reduction Act continued to be allocated to projects moving forward in each State expected to rebuild and modernize infrastructure and strengthen supply chains. Government funds were also awarded to support sustainability, and several cement producers were selected for decarbonization initiatives. Regulators continued to implement clean public procurement strategies and announce research investments. Apparent consumption of cement in 2024 was estimated to be unchanged from that in 2023.

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CEMENT

Company merger-and-acquisition activity in 2024 included the acquisition of a United States-based company's cement plant in Texas by an Ireland-based company and the completed merger of a Colombia-based cement company and a United States-based cement company. In November, a United States-based concrete and cement company reached an agreement to acquire a United States-based cement company, and a Germany-based cement company announced an agreement to acquire another United States-based cement company—each transaction was pending regulatory approval. Plans to increase clinker capacity at a cement plant in Texas and plans to expand a cement plant in Missouri progressed. In May 2024, plans to modernize and expand a cement plant in Wyoming were announced. A cement plant in Indiana was repurposed into a slag-grinding facility in June, and construction of a new slag cement facility in Texas continued. Several minor upgrades were ongoing at some other domestic plants and terminals. Announcements aligned with sustainability included the increased or enhanced use of alternative fuels and materials, carbon capture, utilization and storage projects, improved efficiency, renewable energy, and other innovations. Development of novel cement product lines progressed; two pilot plant cement facilities were planned.

In April 2024, the Connecticut Department of Transportation approved the use of portland-limestone cement (PLC), signifying its adoption by all 50 States and the District of Columbia. Blended cement accounted for 58% of total cement shipments during the first 8 months of 2024, and 97% of the blended shipments were estimated to be PLC (Type IL). In February 2024, the U.S. Environmental Protection Agency issued its "Final Reconsideration of the National Ambient Air Quality Standards for Particulate Matter (PM)"; in March 2024, a coalition of associations expressed concern regarding the lower emissions standard for PM ≤2.5 micrometers in diameter. Many plants have installed emissions-reduction equipment to comply with the 2010 National Emissions Standards for Hazardous Air Pollutants (NESHAP). Some kilns could be shut, idled, or used at reduced capacity to comply with regulations, which would constrain U.S. clinker capacity. In 2022 and 2023, cement plant closures were announced in California, Maine, and New York; in 2024, the plant in Maine transitioned to a distribution center for imported material. Also in 2024, a termination notice was issued to a cement plant in Colorado, and its operational status remained under review by the county.

World Production and Capacity:

	Cement production[e]		Clinker capacity[e]	
	2023	2024	2023	2024
United States (includes Puerto Rico)	90,000	86,000	100,000	100,000
Brazil	67,000	68,000	60,000	60,000
China	2,000,000	1,900,000	2,000,000	1,900,000
Egypt	52,000	50,000	60,000	60,000
India	420,000	450,000	300,000	380,000
Indonesia	67,000	65,000	79,000	79,000
Iran	71,000	72,000	81,000	85,000
Japan	48,000	46,000	54,000	50,000
Korea, Republic of	51,000	52,000	62,000	62,000
Mexico	48,000	48,000	42,000	42,000
Russia	63,000	65,000	80,000	80,000
Saudi Arabia	49,000	50,000	75,000	75,000
Turkey	81,000	82,000	92,000	92,000
Vietnam	110,000	110,000	110,000	110,000
Other countries (rounded)	850,000	860,000	600,000	650,000
World total (rounded)	4,100,000	4,000,000	3,800,000	3,800,000

World Resources: See the Lime and Stone (Crushed) chapters for cement raw-material resources.

Substitutes: Most portland cement is used to make concrete, mortars, or stuccos, and competes in the construction sector with concrete substitutes, such as aluminum, asphalt, clay brick, fiberglass, glass, gypsum (plaster), steel, stone, and wood. Certain materials, especially fly ash and ground granulated blast furnace slag, develop good hydraulic cementitious properties by reacting with lime, such as that released by the hydration of portland cement. Where readily available (including as imports), these SCMs are increasingly being used as partial substitutes for portland cement in many concrete applications and are components of finished blended cements.

[e]Estimated.

[1]Portland cement plus masonry cement unless otherwise specified; excludes Puerto Rico unless otherwise specified.

[2]Includes cement made from imported clinker.

[3]Defined as production of cement (including from imported clinker) + imports (excluding clinker) – exports ± adjustments for stock changes.

[4]Defined as imports (cement and clinker) – exports.

[5]Hydraulic cement and clinker; includes imports into Puerto Rico.

CESIUM

(Data in metric tons, cesium oxide, unless otherwise specified)

Domestic Production and Use: In 2024, no cesium was mined domestically, and the United States was 100% net import reliant for cesium minerals. Pollucite, mainly found in association with lithium-rich, lepidolite-bearing or petalite-bearing zoned granite pegmatites, is the principal cesium ore mineral. Cesium minerals are used as feedstocks to produce a variety of cesium compounds and cesium metal. The primary application for cesium, by gross weight, is in cesium formate brines used for high-pressure, high-temperature well drilling for oil and gas exploration and production. With the exception of cesium formate, cesium is used in relatively small-scale applications, using only a few grams for most applications. Owing to the lack of global availability of cesium, many applications have used mineral substitutes and the use of cesium in any particular application may no longer be viable.

Cesium metal may be used in the production of cesium compounds and photoelectric cells. Cesium bromide may be used in infrared detectors, optics, photoelectric cells, scintillation counters, and spectrophotometers. Cesium carbonate may be used in the alkylation of organic compounds and in energy conversion devices, such as fuel cells, magneto-hydrodynamic generators, and polymer solar cells. Cesium chloride may be used in analytical chemistry applications as a reagent, in high-temperature solders, as an intermediate in cesium metal production, in isopycnic centrifugation, as a radioisotope in nuclear medicine, as an insect repellent in agricultural applications, and in specialty glasses. Cesium hydroxide may be used as an electrolyte in alkaline storage batteries. Cesium iodide may be used in fluoroscopy equipment—Fourier-transform infrared spectrometers—as the input phosphor of X-ray image intensifier tubes, and in scintillators. Cesium nitrate may be used as a colorant and oxidizer in the pyrotechnic industry, in petroleum cracking, in scintillation counters, and in X-ray phosphors. Cesium sulfates may be used in water treatment, fuel cells, and to improve optical quality for scientific instruments.

For industrial uses, cesium catalysts have largely replaced potassium promoters in high-purity sulfuric acid manufacturing, which may enable lower plant stack emissions and lower ignition temperatures. Additionally, cesium catalysts are primarily used in methyl methacrylate manufacturing in place of conventional cyanide-based processes and are necessary to improve efficiency, lower operating costs, and reduce environmental impacts. Sulfuric acid catalysts and methyl methacrylate may be used in aerospace, automotive, and manufacturing applications.

Cesium isotopes, which are obtained as a byproduct in nuclear fission or formed from other isotopes, such as barium-131, may be used in electronic, medical, metallurgical, and research applications. Cesium isotopes are used as an atomic resonance frequency standard in atomic clocks, playing a vital role in aircraft guidance systems, global positioning satellites, and internet and cellular telephone transmissions.

A company in Richland, WA, produced a range of cesium-131 medical products for treatment of various cancers. Cesium-137 may be used in industrial gauges, in mining and geophysical instruments, and for sterilization of food, sewage, and surgical equipment. Because of the danger posed by the radiological properties of cesium-137, Congress set a goal for the National Nuclear Security Administration to eliminate cesium-137 blood irradiators by 2027 in the United States. Alternatives, including X-ray irradiators, have been developed with similar capabilities and have been partially implemented with subsidization.

Salient Statistics—United States: Consumption, import, and export data for cesium have not been available since the late 1980s. Because cesium metal is not traded in commercial quantities, a market price is unavailable. It is estimated that no more than a few thousand kilograms of cesium chemicals are consumed in the United States every year. The United States was 100% net import reliant for its cesium needs, and the primary global producers were estimated to include Canada, China, Germany, and Russia.

In 2024, one company offered 1-gram ampoules of 99.8% (metal basis) cesium for $98.00, a 7% increase from $91.60 in 2023, and 99.98% (metal basis) cesium for $124, a 6% increase from $117 in 2023. At the end of September 2024, the prices for 50 grams of 99.9% (metal basis) cesium acetate, cesium bromide, cesium carbonate, cesium chloride, and cesium iodide were $150.60, $104.00, $135.40, $152.00, and $173.60, respectively, with increases ranging from 5% to 6% compared with prices in 2023.

The price for a cesium-plasma standard solution (10,000 micrograms per milliliter) in 2024 was $93.40 for 50 milliliters and $142.00 for 100 milliliters, increases of 4% from $89.80 and $137.00 in 2023, respectively. The price for 25 grams of 98% (metal basis) cesium formate was $52.40, a 6% increase from $49.40 in 2023.

Recycling: Cesium formate brines are typically rented by oil and gas exploration clients. After completion of the well, the used cesium formate brine is returned and reprocessed for subsequent drilling operations. Cesium formate brines are recycled, recovering nearly 85% of the brines for recycling to be reprocessed for further use.

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CESIUM

Import Sources (2020–23): No reliable data have been available to determine the source of cesium ore imported by the United States since 1988. Prior to 2016, Canada was estimated to be the primary supplier of cesium ore and refined chemicals. Based on recent import data, it was estimated that China and Germany were sources of cesium chemicals.

Tariff: Item	Number	Normal Trade Relations 12–31–24
Alkali metals, other	2805.19.9000	5.5% ad valorem.
Chlorides, other	2827.39.9000	3.7% ad valorem.
Bromides, other	2827.59.5100	3.6% ad valorem.
Iodides, other	2827.60.5100	4.2% ad valorem.
Sulfates, other	2833.29.5100	3.7% ad valorem.
Nitrates, other	2834.29.5100	3.5% ad valorem.
Carbonates, other	2836.99.5000	3.7% ad valorem.

Depletion Allowance: 14% (domestic and foreign).

Government Stockpile: None.

Events, Trends, and Issues: Domestic cesium occurrences will likely remain subeconomic unless market conditions change. No known human health issues are associated with exposure to naturally occurring cesium, and its use has minimal environmental impacts. Manufactured radioactive isotopes of cesium have been known to cause adverse health effects. Certain cesium compounds may be toxic if consumed. Food that has been irradiated using the radioisotope cesium-137 has been found to be safe by the U.S. Food and Drug Administration.

During 2024, one company in Canada reported intermittent cesium production and processing from mined ore and stockpiles at the Tanco Mine. Cesium ore also was estimated to have been mined in China. All other primary mine production of cesium was believed to have ceased within the past two decades. Mining of cesium in Namibia ceased in the early 2000s. The Bikita Mine in Zimbabwe was depleted of pollucite ore reserves in 2018. The Sinclair Mine in Australia completed the mining and shipments of all economically recoverable pollucite ore in 2019.

Throughout 2024, multiple projects that could produce cesium through lepidolite, pollucite, spodumene, and zinnwaldite mining, focused primarily on lithium or cesium extraction, were in the feasibility and exploration stage. One company continued developing a lepidolite concentration mine and processing facility in Namibia, with a targeted lithium hydroxide capacity of 5,700 tons per year expected to commence operations in 2026. Byproduct cesium production was expected to be sent to a downstream chemical conversion facility in Abu Dhabi.

World Mine Production and Reserves:[1] There were no official sources for cesium production data in 2024. Cesium reserves are, therefore, estimated based on the occurrence of pollucite, a primary cesium mineral. Most pollucite contains 5% to 32% cesium oxide. No reliable data were available to determine reserves for specific countries; however, Australia, Canada, China, and Namibia were estimated to have reserves totaling less than 200,000 tons. Existing stockpiles at multiple former mine sites have continued feeding downstream refineries, though recent reports have indicated that stockpiles may be depleted within a few years. The global market of cesium chemical products, excluding cesium formate, was estimated to be 2,200 tons per year. China was estimated to account for 1,000 tons per year of that market. An estimated 11,000 tons of cesium formate were in use, with 5% being depleted and replaced annually.

World Resources:[1] Cesium is associated with lithium-bearing pegmatites worldwide, and cesium resources have been identified in Australia, Canada, Namibia, the United States, and Zimbabwe. In the United States, pollucite occurs in pegmatites in Alaska, Maine, and South Dakota. Lower concentrations occur in brines in Chile and China and in geothermal systems in China, Germany, and India. China was estimated to have cesium-rich deposits of geyserite, lepidolite, and pollucite, with concentrations highest in Yichun, Jiangxi Province, although no resource, reserve, or production estimates were available.

Substitutes: Cesium and rubidium can be used interchangeably in many applications because they have similar physical properties and atomic radii. Cesium, however, is more electropositive than rubidium, making it a preferred material for some applications. However, rubidium is mined from similar deposits, in relatively smaller quantities, as a byproduct of cesium production in pegmatites and as a byproduct of lithium production from lepidolite (hard-rock) mining and processing, making it no more readily available than cesium.

[1]See Appendix C for resource and reserve definitions and information concerning data sources.

CHROMIUM

(Data in thousand metric tons, chromium content, unless otherwise specified)

Domestic Production and Use: In 2024, the United States consumed an estimated 4% of world chromite ore production in various forms of imported materials, such as chromite ore, chromium chemicals, ferrochromium, chromium metal, and stainless steel. Imported chromite ore was consumed by one chemical company to produce chromium chemicals. Stainless-steel and heat-resisting-steel producers were the leading consumers of ferrochromium. Stainless steels and superalloys require the addition of chromium via ferrochromium or chromium-containing scrap. The value of chromium material consumption was estimated to be $900 million in 2024 (as measured by the value of net imports, excluding stainless steel), which was a 6% increase from $846 million in 2023.

Salient Statistics—United States:	2020	2021	2022	2023	2024[e]
Production:					
Mine	—	—	—	—	—
Secondary[1]	119	114	91	126	100
Imports for consumption[2]	448	571	610	451	500
Exports[2]	138	114	132	148	160
Shipments from Government stockpile[3]	5	7	5	NA	NA
Consumption (includes recycling):					
Reported	387	364	275	[e]290	300
Apparent[4]	433	579	574	429	440
Price:[5]					
Chromite ore (gross weight), dollars per metric ton	158	199	277	321	340
Ferrochromium (chromium content), dollars per pound[6]	0.89	1.50	3.19	2.55	1.80
Chromium metal (gross weight), dollars per pound	3.10	4.23	7.20	5.05	5.60
Stocks, consumer, yearend	6	6	5	[e]5	5
Net import reliance[7] as a percentage of apparent consumption	73	80	84	71	77

Recycling: In 2024, recycled chromium (contained in reported stainless-steel scrap receipts) accounted for 23% of apparent consumption.

Import Sources (2020–23): Chromite (ores and concentrates): South Africa, 96%; Turkey, 3%; and other, 1%.
Chromium-containing scrap:[8] Canada, 51%; Mexico, 43%; and other, 6%.
Chromium (primary metal):[9] South Africa, 25%; Kazakhstan, 14%; Finland, 7%; Russia, 6%; and other, 48%.
Chromium-containing chemicals: Kazakhstan, 24%; China,[10] 18%; Germany, 17%; Italy, 12%; and other, 29%.
Total imports: South Africa, 32%; Kazakhstan, 11%; Canada, 6%; Finland, 6%; and other, 45%.

Tariff: Item	Number	Normal Trade Relations 12–31–24
Chromium ores and concentrates:		
Not more than 40% chromic oxide (Cr_2O_3)	2610.00.0020	Free.
More than 40% but less than 46% Cr_2O_3	2610.00.0040	Free.
More than or equal to 46% Cr_2O_3	2610.00.0060	Free.
Ferrochromium:		
More than 4% carbon	7202.41.0000	1.9% ad valorem.
More than 3% but less than 4% carbon	7202.49.1000	1.9% ad valorem.
More than 0.5% but less than 3% carbon	7202.49.5010	3.1% ad valorem.
Not more than 0.5% carbon	7202.49.5090	3.1% ad valorem.
Ferrosilicon chromium	7202.50.0000	10% ad valorem.
Stainless-steel scrap	7204.21.0000	Free.
Chromium metal:		
Unwrought, powder	8112.21.0000	3% ad valorem.
Waste and scrap	8112.22.0000	Free.
Other	8112.29.0000	3% ad valorem.

Depletion Allowance: 22% (domestic), 14% (foreign).

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CHROMIUM

Government Stockpile (gross weight):[11]

Material	FY 2024 Potential acquisitions	FY 2024 Potential disposals	FY 2025 Potential acquisitions	FY 2025 Potential disposals
Ferrochromium[12]	—	21.8	—	21.8
Chromium metal	—	0.454	—	0.454

Events, Trends, and Issues: South Africa was the leading chromite ore producer. Global chromite ore mine production was estimated to have increased by 4 in% 2024 compared with production in 2023. Production in South Africa, the world's leading producer of chromite, increased by an estimated 7% compared with production in 2023, largely owing to an increase in the average price of chromite ore. However, challenges related to deep-level mining, increased labor costs, ongoing issues with rail transportation, and an unreliable supply of electricity could affect production in South Africa.

China was the leading ferrochromium- and stainless-steel-producing country and the leading chromium-consuming country. However, the production of stainless steel in China has been affected by oversupply, decreases in consumer demand, and escalating trade tensions, which have led to decreases in the price of ferrochromium.

World Mine Production (gross weight) and Reserves: Reserves for Kazakhstan were revised based on a Government report.

	Mine production (marketable) 2023	Mine production (marketable) 2024[e]	Reserves[13] (shipping grade)[14]
United States	—	—	630
Brazil	1,420	1,400	6,600
Finland	1,910	1,900	8,300
India[e]	4,100	4,100	79,000
Kazakhstan[e]	6,000	6,500	320,000
South Africa	19,700	21,000	200,000
Turkey	8,160	8,000	27,000
Zimbabwe	1,070	1,100	540,000
Other countries	2,880	2,900	NA
World total (rounded)	45,200	47,000	>1,200,000

World Resources:[13] World resources are greater than 12 billion tons of shipping-grade chromite, sufficient to meet conceivable demand for centuries. World chromium resources are heavily geographically concentrated (95%) in Kazakhstan and southern Africa; United States chromium resources are mostly in the Stillwater Complex in Montana.

Substitutes: Chromium has no substitute in stainless steel, the leading end use, or in superalloys, the major strategic end use. Chromium-containing scrap can substitute for ferrochromium in some metallurgical uses.

[e]Estimated. NA Not available. — Zero.

[1]Secondary production is based on reported receipts of all types of stainless-steel scrap.

[2]Includes chromium chemicals, chromium metal, chromite ores, ferrochromium, ferrosilicon chromium, and stainless-steel products and scrap.

[3]Defined as change in total inventory from prior yearend inventory. Beginning in 2023, Government stock changes no longer available.

[4]Defined for 2020–22 as production (from mines and secondary) + imports – exports ± adjustments for Government and industry stock changes. Beginning in 2023, Government stock changes no longer included.

[5]Source: Argus Media Group, Argus Non-Ferrous Markets.

[6]Excludes ferrosilicon chromium.

[7]Defined for 2020–22 as imports – exports ± adjustments for Government and industry stock changes. Beginning in 2023, Government stock changes no longer included.

[8]Includes chromium metal scrap and stainless-steel scrap.

[9]Includes chromium metal, ferrochromium, and stainless steel.

[10]Includes Hong Kong.

[11]See Appendix B for definitions.

[12]High-carbon and low-carbon ferrochromium, combined.

[13]See Appendix C for resource and reserve definitions and information concerning data sources.

[14]Units are thousand metric tons gross weight of shipping-grade chromite ore, which is deposit quantity and grade normalized to 45% Cr_2O_3, except for the United States, where grade is normalized to 7% Cr_2O_3, and Finland, where grade is normalized to 26% Cr_2O_3.

CLAYS

(Data in thousand metric tons unless otherwise specified)

Domestic Production and Use: Production of clays (sold or used) in the United States was estimated to be 26 million tons valued at $1.7 billion in 2024, with about 120 companies operating clay and shale mines in 38 States. The leading 20 companies produced approximately 66% of the U.S. tonnage and 82% of the value for all types of clay. Principal domestic uses for specific clays were estimated to be as follows: ball clay (61% floor and wall tile), bentonite (48% pet waste absorbents and 23% drilling mud), common clay (47% brick, 25% lightweight aggregate, and 22% cement), fuller's earth (77% absorbents, including oil and grease absorbents, pet waste absorbents, and miscellaneous absorbents), and kaolin (54% fillers, extenders, and binders and 23% ceramics). Fire clay uses were withheld to avoid disclosing company proprietary data.

In 2024, the United States exported an estimated 700,000 tons of bentonite; Canada, Japan, and Mexico, in decreasing order, were the leading destinations. About 1.6 million tons of kaolin was exported mainly as a paper coating and filler; a component in ceramic bodies; and fillers and extenders in paint, plastic, and rubber products; China, Mexico, and Japan, in decreasing order, were the leading destinations. Lesser quantities of ball clay, fire clay, and fuller's earth were exported.

Salient Statistics—United States:	**2020**	**2021**	**2022**	**2023**	**2024[e]**
Production (sold or used):					
Ball clay	985	[e]1,080	[e]1,030	[e]1,000	1,000
Bentonite	4,250	4,580	4,580	4,360	4,800
Common clay	12,900	12,700	12,900	12,500	13,000
Fire clay	635	675	[e]622	685	670
Fuller's earth[e, 1]	2,120	2,240	2,200	2,260	2,400
Kaolin[e]	4,640	4,360	4,340	4,560	4,500
Total[1, 2]	25,500	25,700	25,700	25,400	26,000
Imports for consumption:					
Artificially activated clays and earths	31	41	58	72	66
Kaolin	224	149	200	125	130
Other	28	47	49	35	66
Total[2]	284	237	306	232	270
Exports:					
Artificially activated clays and earths	127	139	134	92	92
Ball clay	68	139	165	145	170
Bentonite	728	861	830	785	700
Clays, not elsewhere classified	185	186	208	194	180
Fire clay[3]	190	210	158	133	140
Fuller's earth	77	83	87	70	75
Kaolin	1,990	2,330	2,020	1,510	1,600
Total[2]	3,360	3,950	3,610	2,930	2,900
Consumption, apparent[4]	22,400	22,000	22,400	22,700	23,000
Price, average unit value, ex-works, dollars per metric ton:					
Ball clay	46	46	47	44	44
Bentonite	97	100	101	102	99
Common clay	17	17	17	18	18
Fire clay	12	12	12	15	15
Fuller's earth[1]	90	88	91	91	90
Kaolin	159	152	157	161	160
Employment (excludes office workers), number:[e]					
Mine (may not include contract workers)	1,060	1,060	1,060	1,110	1,200
Mill	4,260	4,240	4,240	4,320	4,400
Net import reliance[5] as a percentage of apparent consumption	E	E	E	E	E

Recycling: Insignificant.

Import Sources (2020–23): All clay types combined: Brazil, 62%; Mexico, 21%; China, 3%; Spain 3%; and other, 11%.

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CLAYS

Tariff: Item	Number	Normal Trade Relations 12–31–24
Kaolin and other kaolinic clays, whether or not calcined	2507.00.0000	Free.
Bentonite	2508.10.0000	Free.
Fire clay	2508.30.0000	Free.
Common blue clay and other ball clays	2508.40.0110	Free.
Decolorizing earths and fuller's earth	2508.40.0120	Free.
Other clays	2508.40.0150	Free.
Chamotte or dinas earth	2508.70.0000	Free.
Activated clays and activated earths	3802.90.2000	2.5% ad valorem.
Expanded clays and other mixtures	6806.20.0000	Free.

Depletion Allowance: Ball clay, bentonite, fire clay, fuller's earth, and kaolin, 14% (domestic and foreign); clay used in the manufacture of common brick, lightweight aggregate, and sewer pipe, 7.5% (domestic and foreign); clay used in the manufacture of drain and roofing tile, flowerpots, and kindred products, 5% (domestic and foreign); clay from which alumina and aluminum compounds are extracted, 22% (domestic).

Government Stockpile: None.

Events, Trends, and Issues: The total tonnage of clays sold or used by domestic producers increased from that in 2023; ball clay, bentonite, common clay, and fuller's earth, increased or were unchanged, whereas fire clay and kaolin decreased in 2024. Imports for all types of clay increased by 14% to 270,000 tons; Brazil and Mexico, in decreasing order, were the major sources for imported clays in 2024. U.S. apparent consumption in 2024 was estimated to be 23 million tons, compared with 22.7 million tons in 2023.

World Mine Production and Reserves:[6] Global reserves are large, but country-specific data were not available.

	Mine production					
	Bentonite		Fuller's earth		Kaolin	
	2023	2024[e]	2023	2024[e]	2023	2024[e]
United States	4,360	4,800	[1]2,260	[1]2,400	[e]4,560	4,500
Brazil (beneficiated)	386	390	—	—	828	830
China	[e]2,100	2,100	—	—	[e]7,800	7,800
Czechia	196	200	—	—	[7]2,400	[7]2,400
Denmark	[e]925	930	—	—	—	—
Greece	[7]1,110	[7]1,100	49	50	—	—
India	[e]3,700	3,700	[e]730	730	[e, 7]8,400	[7]8,400
Iran	[e]850	850	—	—	[e]2,100	2,100
Mexico	[e]79	80	[e]120	120	[e]230	230
Russia	[e]36	40	—	—	[e]2,500	2,500
Senegal	—	—	[e]150	150	—	—
Spain	118	120	[e]620	620	[e, 7]280	[7]280
Turkey	2,490	2,500	74	70	1,350	1,300
Uzbekistan	[e]60	60	—	—	[e]4,000	4,000
Other countries	3,870	3,900	195	190	9,950	9,900
World total (rounded)[2]	20,300	21,000	[1]4,200	[1]4,300	44,400	44,000

World Resources:[6] Resources of all clays are extremely large.

Substitutes: Clays compete with calcium carbonate in filler and extender applications; diatomite, organic pet litters, polymers, silica gel, and zeolites as absorbents; and various siding and roofing types in building construction.

[e]Estimated. E Net exporter. — Zero.

[1]Does not include U.S. production of attapulgite.

[2]Data may not add to totals shown because of independent rounding.

[3]Includes refractory-grade kaolin.

[4]Defined as production (sold or used) + imports – exports.

[5]Defined as imports – exports.

[6]See Appendix C for resource and reserve definitions and information concerning data sources.

[7]Includes production of crude ore.

COBALT

(Data in metric tons, cobalt content, unless otherwise specified)

Domestic Production and Use: In 2024, the Eagle Mine, a nickel-copper mine in Michigan, produced cobalt-bearing nickel concentrate, which was exported to Canada or overseas for processing. Mining activity at a cobalt-copper-gold mine in Idaho remained suspended in 2024 owing to low cobalt prices. Most U.S. cobalt supply consisted of imports and secondary (scrap) materials. About five companies in the United States produced cobalt chemicals. An estimated 51% of cobalt consumed in the United States was used in superalloys, mainly aircraft gas turbine engines; 25% in a variety of chemical applications; 15% in various other metallic applications; and 9% in cemented carbides for cutting and wear-resistant applications. The total estimated value of cobalt consumed in 2024 was $260 million.

Salient Statistics—United States:	2020	2021	2022	2023	2024[e]
Production:[e]					
Mine	600	650	500	500	300
Secondary[1]	2,010	1,800	1,920	2,030	2,000
Imports for consumption	9,740	9,790	10,500	9,500	11,000
Exports	3,430	4,930	5,360	5,110	4,500
Consumption (includes secondary):					
Estimated[2]	7,260	7,270	7,570	7,840	8,000
Apparent[e, 3]	8,480	6,650	7,150	6,440	8,500
Price, average, dollars per pound:					
U.S. spot, cathode[4]	15.70	24.21	30.78	17.20	17
London Metal Exchange (LME), cash	14.21	23.17	28.83	15.48	12
Stocks, yearend:					
Industry[e, 2, 5]	952	1,010	946	925	900
LME, U.S. warehouse	82	50	34	34	34
Net import reliance[6] as a percentage of apparent consumption	76	73	73	69	76

Recycling: In 2024, cobalt content of purchased scrap represented 25% of estimated cobalt consumption.

Import Sources (2020–23): Metal, oxide, and salts: Norway, 27%; Finland, 17%; Japan, 14%; Canada, 13%; and other, 29%.

Tariff: Item	Number	Normal Trade Relations 12–31–24
Cobalt ores and concentrates	2605.00.0000	Free.
Chemical compounds:		
Cobalt oxides and hydroxides	2822.00.0000	0.1% ad valorem.
Cobalt chlorides	2827.39.6000	4.2% ad valorem.
Cobalt sulfates	2833.29.1000	1.4% ad valorem.
Cobalt carbonates	2836.99.1000	4.2% ad valorem.
Cobalt acetates	2915.29.3000	4.2% ad valorem.
Unwrought cobalt, alloys	8105.20.3000	4.4% ad valorem.
Unwrought cobalt, other	8105.20.6000	Free.
Cobalt mattes and other intermediate products; cobalt powders	8105.20.9000	Free.
Cobalt waste and scrap	8105.30.0000	Free.
Wrought cobalt and cobalt articles	8105.90.0000	3.7% ad valorem.

Depletion Allowance: 22% (domestic), 14% (foreign).

Government Stockpile:[7]

	FY 2024		FY 2025	
Material	Potential acquisitions	Potential disposals	Potential acquisitions	Potential disposals
Cobalt alloys, gross weight[8]	200	—	60	—

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COBALT

Events, Trends, and Issues: Global cobalt mine and refinery production were estimated to have increased to another record high in 2024. The increase in mine production was mainly in Congo (Kinshasa), the world's leading source of mined cobalt, which accounted for an estimated 76% of world cobalt mine production, followed by Indonesia, which accounted for 10%. China was the world's leading producer of refined cobalt and increased metal refining capacity throughout the year. The majority of China's refinery production was from partially refined cobalt imported from Congo (Kinshasa) and Indonesia. China was the world's leading consumer of cobalt, with the majority used by the lithium-ion battery industry. New production of mined and refined cobalt has led to excess global supply and lower cobalt prices. In 2024, the United States enacted tariff rate increases on cobalt ores and concentrates originating from China, as well as cobalt-containing products including electric vehicles and lithium-ion batteries.

World Mine Production and Reserves: Reserves for the United States, Canada, Indonesia, Papua New Guinea, and "Other countries" were revised based on company and Government reports.

	Mine production[e]		Reserves[9]
	2023	2024	
United States	500	300	70,000
Australia	5,220	3,600	[10]1,700,000
Canada	4,220	4,500	220,000
Congo (Kinshasa)	175,000	220,000	6,000,000
Cuba	3,300	3,500	500,000
Indonesia	19,000	28,000	640,000
Madagascar	4,000	2,600	100,000
New Caledonia[11]	2,570	1,500	NA
Papua New Guinea	3,070	2,800	62,000
Philippines	3,800	3,800	260,000
Russia	8,700	8,700	250,000
Turkey	2,500	2,700	91,000
Other countries	6,080	6,200	800,000
World total (rounded)	238,000	290,000	11,000,000

World Resources:[9] Identified cobalt resources of the United States are estimated to be about 1 million tons. Most of these resources are in Minnesota. Other notable occurrences are in Alaska, California, Idaho, Michigan, Missouri, Montana, Oregon, and Pennsylvania. Identified world terrestrial cobalt resources are about 25 million tons. The vast majority of global resources are in sediment-hosted stratiform copper deposits in Congo (Kinshasa) and Zambia; nickel-bearing laterite deposits in Australia and nearby island countries and Cuba; and magmatic nickel-copper sulfide deposits of mafic and ultramafic rocks in Australia, Canada, Russia, and the United States.

Substitutes: Depending on the application, substitution for cobalt could result in a loss in product performance or increase cost. The cobalt contents of lithium-ion batteries, the leading global use for cobalt, are being reduced; cobalt-free substitutes that use iron and phosphorus held significant market share in China. Potential substitutes in other applications include barium or strontium ferrites, neodymium-iron-boron alloys, or nickel-iron alloys in magnets; cerium, iron, lead, manganese, or vanadium in paints; cobalt-iron-copper or iron-copper in diamond tools; copper-iron-manganese for curing unsaturated polyester resins; iron, iron-cobalt-nickel, nickel, ceramic-metallic composites (cermets), or ceramics in cutting and wear-resistant materials; nickel-base alloys or ceramics in jet engines; nickel in petroleum catalysts; rhodium in hydroformylation catalysts; and titanium-base alloys in prosthetics.

[e]Estimated, — Zero. NA Not available.

[1]Estimated from consumption of purchased scrap.

[2]Includes reported data and U.S. Geological Survey estimates.

[3]Defined for 2020–22 as secondary production + imports – exports ± adjustments for Government and industry stock changes for refined cobalt. Beginning in 2023, Government stock changes no longer included.

[4]Source: S&P Global Platts Metals Week. Cobalt cathode is refined cobalt metal produced by an electrolytic process.

[5]Stocks held by consumers and processors; excludes stocks held by trading companies and held for investment purposes.

[6]Defined for 2020–22 as imports – exports ± adjustments for Government and industry stock changes for refined cobalt. Beginning in 2023, Government stock changes no longer included.

[7]See Appendix B for definitions.

[8]Samarium-cobalt alloy; excludes potential disposals of aerospace alloys.

[9]See Appendix C for resource and reserve definitions and information concerning data sources.

[10]For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 610,000 tons.

[11]Overseas territory of France.

COPPER

(Data in thousand metric tons, copper content, unless otherwise specified)

Domestic Production and Use: In 2024, the recoverable copper content of U.S. mine production was an estimated 1.1 million tons, a decrease of 3% from that in 2023, and was valued at an estimated $10 billion, slightly greater than $9.83 billion in 2023. Arizona was the leading copper-producing State and accounted for approximately 70% of domestic output; copper was also mined in Michigan, Missouri, Montana, Nevada, New Mexico, and Utah. Copper was recovered or processed at 25 mines (17 of which accounted for more than 99% of mine production), 2 primary smelters, 1 secondary smelter, 2 primary electrolytic refineries, 14 electrowon refineries, and 3 secondary fire refineries. A new secondary smelter and secondary refinery were expected to start up by yearend. Refined copper and scrap were consumed at about 30 brass mills, 14 rod mills, and several hundred foundries and miscellaneous manufacturers. According to the Copper Development Association, copper and copper alloy products were used in building construction, 42%; electrical and electronic products, 23%; transportation equipment, 18%; consumer and general products, 10%; and industrial machinery and equipment, 7%.

Salient Statistics—United States:	**2020**	**2021**	**2022**	**2023**	**2024[e]**
Production:					
Mine, recoverable	1,200	1,230	1,230	1,130	1,100
Refinery:					
Primary (from ore)	872	922	930	843	850
Secondary (from scrap)	43	49	40	39	40
Copper recovered from old (post-consumer) scrap[1]	161	169	152	[e]150	150
Imports for consumption:					
Ore and concentrates	2	11	12	3	([2])
Refined	676	919	732	771	810
Exports:					
Ore and concentrates	383	344	351	339	320
Refined	41	48	27	33	60
Consumption:					
Reported, refined copper	1,680	1,750	1,720	1,570	1,600
Apparent, primary refined copper and copper from old scrap[3]	1,660	1,960	1,820	1,690	1,800
Price, annual average, cents per pound:					
U.S. producer, cathode (COMEX + premium)	286.7	432.3	410.8	395.3	430
COMEX, high-grade, first position	279.9	424.3	400.7	385.7	420
London Metal Exchange, grade A, cash	279.8	422.5	399.8	384.8	420
Stocks, refined, held by U.S. producers, consumers, and metal exchanges, yearend	118	117	84	127	70
Employment, mine and plant, number	11,000	11,400	12,000	12,600	13,000
Net import reliance[4] as a percentage of apparent consumption	38	44	41	41	45

Recycling: Old (post-consumer) scrap, converted to refined metal, alloys, and other forms, provided an estimated 150,000 tons of copper in 2024, and an estimated 720,000 tons of copper was recovered from new (manufacturing) scrap derived from fabricating operations. Brass and wire-rod mills accounted for approximately 85% of the total copper recovered from scrap. Copper recovered from scrap contributed about 35% of the U.S. copper supply.[5]

Import Sources (2020–23): Copper content of blister and anodes: Finland, 92%; Malaysia, 3%; and other, 5%. Copper content of matte, ash, and precipitates: Canada, 48%; Belgium, 23%; Japan, 13%; Spain, 6%; and other, 10%. Copper content of ore and concentrates: Canada, >99%; and other, <1%. Copper content of scrap: Canada, 46%; Mexico, 42%; Dominican Republic, 3%; and other, 9%. Refined copper: Chile, 65%; Canada, 17%; Mexico, 9%; Peru, 6%; and other, 3%. Refined copper accounted for 88% of all unmanufactured copper imports.

Tariff: Item	**Number**	**Normal Trade Relations 12–31–24**
Copper ore and concentrates, copper content	2603.00.0010	1.7¢/kg on lead content.
Unrefined copper anodes	7402.00.0000	Free.
Refined copper and alloys, unwrought	7403.00.0000	1% ad valorem.
Copper scrap	7404.00.0000	Free.
Copper wire rod	7408.11.0000	1% or 3% ad valorem.

Depletion Allowance: 15% (domestic), 14% (foreign).

Government Stockpile: None.

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COPPER

Events, Trends, and Issues: In 2024, production decreased at a majority of copper mines in the United States, and domestic mined copper output declined by an estimated 3% from that in 2023. At the Bingham Canyon Mine in Utah, changes to the mine plan required to mitigate geotechnical risks resulted in lower ore grades and copper recoveries. Production at the Eagle Mine in Michigan was affected by decreased copper ore grades and reduced mill throughput rates owing to a fall of ground along an ore access ramp. Output also decreased at multiple mines in Arizona and New Mexico because of lower ore grades and mining rates. These decreases were partially offset by a significant increase in mined copper production at the Robinson Mine in Nevada owing to planned mine sequencing that yielded higher ore grades and copper recovery rates. At U.S. refineries, copper production increased slightly in 2024 compared with that in 2023. The Kennecott smelter and electrolytic refinery near Salt Lake City, UT, returned to normal operations in the first quarter of 2024 following major rebuilds in 2023. A new secondary copper refinery in Kentucky and a new secondary copper smelter in Georgia were expected to begin operating by yearend 2024.

The COMEX copper price reached a record high in May 2024 and was projected to average $4.20 per pound in full year 2024, an increase of 9% from the annual average price in 2023. Analysts attributed the higher price to multiple factors, such as expectations for reduced global copper supply in the near future, optimistic sentiment regarding world copper demand, strong manufacturing production in China, and decreasing inflation in the United States.

World Mine and Refinery Production and Reserves: Reserves for Canada, Indonesia, Peru, and the United States were revised based on company, Government, and (or) industry association reports.

	Mine production		Refinery production		Reserves[6]
	2023	2024[e]	2023	2024[e]	
United States	1,130	1,100	882	890	47,000
Australia	778	800	442	460	[7]100,000
Canada	500	450	315	320	8,300
Chile	5,250	5,300	2,080	1,900	190,000
China	1,820	1,800	12,000	12,000	41,000
Congo (Kinshasa)	2,930	3,300	2,170	2,500	80,000
Germany	—	—	609	630	—
India	27	30	509	510	2,200
Indonesia	907	1,100	225	350	21,000
Japan	—	—	1,490	1,600	—
Kazakhstan	[e]740	740	458	470	20,000
Korea, Republic of	—	—	604	620	—
Mexico	699	700	509	350	53,000
Peru	2,760	2,600	403	390	100,000
Poland	395	410	592	590	34,000
Russia	[e]890	930	[e]1,000	960	80,000
Zambia	712	680	222	170	21,000
Other countries	3,020	2,700	2,460	2,500	180,000
World total (rounded)	22,600	23,000	27,000	27,000	980,000

World Resources:[6] The most recent U.S. Geological Survey assessment of global copper resources indicated that, as of 2015, identified resources contained 1.5 billion tons of unextracted copper (2.1 billion tons when past production of 0.6 billion tons is included) and undiscovered resources contained an estimated 3.5 billion tons of copper.[8]

Substitutes: Aluminum substitutes for copper in automobile radiators, cooling and refrigeration tube, electrical equipment, and power cable. Optical fiber substitutes for copper in telecommunications applications, and plastics substitute for copper in drain pipe, plumbing fixtures, and water pipe. Titanium and steel are used in heat exchangers.

[e]Estimated. — Zero.

[1]Copper converted to refined metal, alloys, and other forms by brass and wire-rod mills, foundries, refineries, and other manufacturers.

[2]Less than ½ unit.

[3]Primary refined production + copper recovered from old scrap + refined imports – refined exports ± adjustments for refined copper stock changes.

[4]Defined as refined imports – refined exports ± adjustments for refined copper stock changes.

[5]Primary refined production + copper from old and new scrap + refined imports – refined exports ± adjustments for refined copper stock changes.

[6]See Appendix C for resource and reserve definitions and information concerning data sources.

[7]For Australia, Joint Ore Reserves Committee-compliant or equivalent reserves were 27 million tons.

[8]Source: Hammarstrom, J.M., Zientek, M.L., Parks, H.L., Dicken, C.L., and the U.S. Geological Survey Global Copper Mineral Resource Assessment Team, 2019, Assessment of undiscovered copper resources of the world, 2015 (ver. 1.2, December 2021): U.S. Geological Survey Scientific Investigations Report 2018–5160, 619 p. (Accessed November 18, 2024, at https://doi.org/10.3133/sir20185160.)